5. Propagarea campului electromagnetic

(neghidata/ghidata)

SUPORT DIDACTIC PENTRU DISCIPLINA – "FIZICA III – ELECTROMAGNETISM"

PREDATA STUDENTILOR DE LA FACULTATEA DE AUTOMATICA SI CALCULATOARE, 2022-2023

UNIVERSITATEA POLITEHNICA BUCURESTI

PROF. *GABRIELA CIUPRINA*

S.L. *SORIN LUP*

Cuprins

- 5.0. Introducere
- 5.1. Propagarea undelor EM in dielectrici cu pierderi (regim armonic permanent r.a.p.)
- 5.2. Unda plana
- 5.3. Unda plana. Propagarea in medii nemarginite
- 5.4. Unda plana. Reflexii si refractii. (Incidenta normala)
- 5.5. Polarizarea undelor.
- 5.6. Reflexii si refractii – incidenta oblica
- 5.7. Puterea transmisa
- 5.8. Ghiduri de unda (facultativ)
- 5.9. Sisteme rezonante (facultativ)

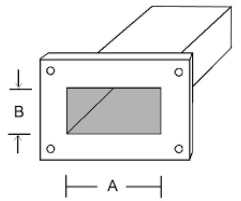
5.0. Introducere

In capitolul anterior a fost studiata propagarea campului electromagnetic, **ghidat de linii** electrice. In ipotezele explicate la modelulul liniei, chiar daca regimul campului EM este general variabil, datorita neglijarii componentelor longitudinale ale curentului de deplasare in lungul liniei, in plane transversale se pastreaza relatiile din regim cvasistationar. Aceasta a permis explicarea proprietatilor liniei cu o aproximatie suficienta din punct de vedere a teoriei circuitelor electrice. Intreaga linie poate fi inteleasa ca un circuit electric cu parametri distribuiti, caracterizabila local prin parametri lineici (raportati la unitatea de lungime). Propagarea ghidata de-a lungul liniilor electrice este cel mai simplu mod de propagare a campului EM, ea poate fi modelata cu circuite cu parametri distribuiti si simulata in simulatoare de circuit, de tip SPICE.

In acest capitol vom studia propagarea campului EM in general:

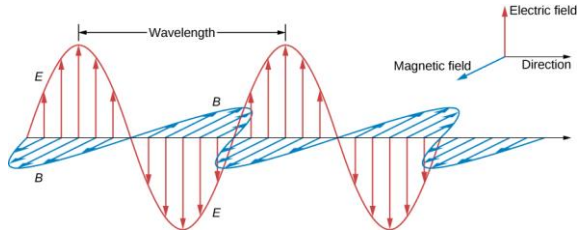
- Neghidata (in intreg spatiul)
- Ghidata de “ghiduri de unda”(dar nu linii)



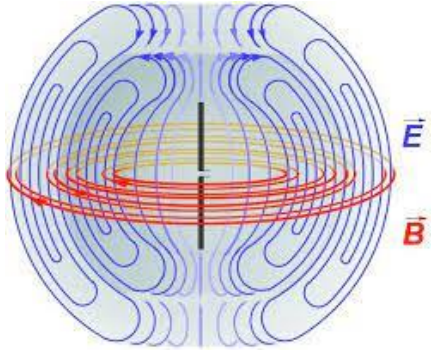


5.0. Introducere

In general, probleme de analiza a campului EM in regim general variabil sunt



- **Probleme de propagare** – in domenii nu exista surse interne de camp, campul EM se propaga in spatiul liber sau in domenii marginite de frontiere metalice (numite dispozitive de radio frecventa-RF sau de microunde-MW cum sunt cupoarele, mixerele, spliterele, filtrele), datorita unor conditii initiale in domeniu, sau pe frontiera lui. Intereseaza modul in care campul se propaga (cap.5, nu linii).



- **Probleme de radiatie** – in domenii exista surse de camp – antene (cap.6), sau imprastierea undelor de diferite obiecte (de exemplu in radar), dar care nu fac obiectul acestui curs.

5.1. Propagarea undelor EM in dielectrici cu pierderi

- **Dielectric cu pierderi** = un dielectric in care unda EM isi pierde din putere pe masura ce se propaga, datorita de exemplu unor proprietati de conductie (slabe) ale mediului. $\sigma \neq 0$.
- **Dielectric fara pierderi**: $\sigma = 0$

Ipoteze de studiu: 1) mediu liniar, omogen, izotrop, neelectrizat

Materiale liniare din toate punctele de vedere: electric (ε), magnetic (μ), conductie (σ)

Proprietatile de material nu depind de punctul din spatiu, sunt aceleasi in toate punctele domeniului de stiudiu.

Proprietatile de material nu depend de directia campului. Constantele de material sunt scalari, nu tensori.

$\rho = 0$

2) Vom presupune un regim armonic permanent (r.a.p), de pulsatie ω

→ Pentru scrierea relatiilor matematice vom folosi reprezentarea in complex.

5.1. Propagarea undelor EM in dielectrici cu pierderi

Reamintim ecuatile Maxwell – formele locale ale legilor:

$$\operatorname{div}\vec{D} = \rho_v \qquad \nabla\cdot\vec{D} = \rho_v$$

$$\operatorname{div}\vec{B} = 0 \qquad \nabla\cdot\vec{B} = 0$$

$$\operatorname{rot}\vec{E} = -\frac{\partial\vec{B}}{\partial t} \equiv \nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\operatorname{rot}\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t} \qquad \nabla\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$$

$$\vec{D} = \bar{\bar{\varepsilon}}\vec{E} + \vec{P}_p \qquad \vec{D} = \bar{\bar{\varepsilon}}\vec{E} + \vec{P}_p$$

$$\vec{B} = \bar{\bar{\mu}}\vec{H} + \mu_0\vec{M}_p \qquad \vec{B} = \bar{\bar{\mu}}\vec{H} + \mu_0\vec{M}_p$$

$$\vec{J} = \bar{\bar{\sigma}}\left(\vec{E} + \vec{E}_i\right) \qquad \vec{J} = \bar{\bar{\sigma}}\left(\vec{E} + \vec{E}_i\right)$$

In ipotezele mentionate (materiale liniare, regim armonic permanent)

$$\nabla\cdot\underline{\vec{D}} = 0$$

$$\nabla\cdot\underline{\vec{B}} = 0$$

$$\nabla\times\underline{\vec{E}} = -j\omega\underline{\vec{B}}$$

$$\nabla\times\underline{\vec{H}} = \underline{\vec{J}} + j\omega\underline{\vec{D}}$$

$$\underline{\vec{D}} = \varepsilon\underline{\vec{E}}$$

$$\underline{\vec{B}} = \mu\underline{\vec{H}}$$

$$\underline{\vec{J}} = \sigma\underline{\vec{E}}$$

Liniile ambelor campuri E si H sunt curbe inchise. Cele doua campuri tind sa fie ortogonale unul pe celalalt (H este de-a lungul axei liniei lui E si invers). Daca stim unul din campuri, celalalt este usor de calculat.

In ipotezele mentionate (materiale omogene)

$$\nabla\cdot\underline{\vec{D}} = \nabla\cdot\left(\varepsilon\underline{\vec{E}}\right) = \varepsilon\,\nabla\cdot\underline{\vec{E}}$$

$$\nabla\cdot\underline{\vec{B}} = \nabla\cdot\left(\mu\underline{\vec{H}}\right) = \mu\,\nabla\cdot\underline{\vec{H}}$$

$$\nabla\cdot\underline{\vec{E}} = 0$$

$$\nabla\cdot\underline{\vec{H}} = 0$$

$$\nabla\times\underline{\vec{E}} = -j\omega\mu\underline{\vec{H}}$$

$$\nabla\times\underline{\vec{H}} = (\sigma + j\omega\varepsilon)\underline{\vec{E}}$$

Obs:

Deoarece $\operatorname{div}\operatorname{rot} \equiv 0$ ➡ $\nabla\cdot\nabla\times\underline{\vec{H}} = 0$,

$\nabla\cdot(\sigma + j\omega\varepsilon)\underline{\vec{E}} = 0$, $\nabla\cdot\underline{\vec{E}} = 0$, $\nabla\cdot\vec{D} = 0$, ➡ $\rho_v = 0$

Ipoteza $\rho = 0$ nu era necesara, sarcina se relaxeaza in r.a.p.

5.1. Propagarea undelor EM in dielectrici cu pierderi

Ecuatii de ordinul I

$$\nabla \cdot \underline{\vec{E}} = 0$$

$$\nabla \cdot \underline{\vec{H}} = 0$$

$$\nabla \times \underline{\vec{E}} = -j\omega\mu\underline{\vec{H}}$$

$$\nabla \times \underline{\vec{H}} = (\sigma + j\omega\varepsilon)\underline{\vec{E}}$$

Un material dielectric nu se polarizeaza instantaneu (exista un histerezis dielectric), ceea ce reflecta (in r.a.p) intr-o legatura de material D-E in care permitivitatea este un numar complex.

$\underline{\vec{D}} = \underline{\varepsilon}\,\underline{\vec{E}}$ unde $\underline{\varepsilon} = \varepsilon' - j\,\varepsilon''$

ε'- are aceeasi semnificatie ca cea discutata pana acum, ea este cea folosinta in regim ES pentru calculul capacitatilor.

ε''- caracterizeaza pierderile in dielectric datorita polarizarii materialului.

Se defineste unghiul de pierderi (prin histerezis dielectric) δ_h unde $\operatorname{tg}\delta_h = \frac{\varepsilon''}{\varepsilon'}$

$$\nabla \times \underline{\vec{H}} = (\sigma + j\omega\underline{\varepsilon})\underline{\vec{E}} = (\sigma + \omega\varepsilon'' + j\omega\varepsilon')\underline{\vec{E}} = (\sigma' + j\omega\varepsilon)\underline{\vec{E}}$$

$$\operatorname{tg}\delta = \frac{\sigma'}{\omega\varepsilon'} = \frac{\sigma}{\omega\varepsilon'} + \frac{\varepsilon''}{\varepsilon'} = \operatorname{tg}\delta_c + \operatorname{tg}\delta_h$$

unde $\sigma' = \sigma + \omega\varepsilon''$

Permitivitatea imaginara (care descrie perderile din dielectric) se modeleaza deci in r.a.p. ca o coductivitate suplimentara.

$\operatorname{tg}\delta_c$: factor de pierderi prin conductie

$\operatorname{tg}\delta_h$: factor de pierderi prin histerezis dielectric

In cele ce urmeaza presupunem ε ca avand semnificatia din ES, si eventualele pierderi datorate polarizarii incluse in valoarea conductivitatii σ

5.1. Propagarea undelor EM in dielectrici cu pierderi

Ecuatii de ordinul I

$$\nabla \cdot \underline{\vec{E}} = 0 \quad (1)$$

$$\nabla \cdot \underline{\vec{H}} = 0 \quad (2)$$

$$\nabla \times \underline{\vec{E}} = -j\omega\mu \underline{\vec{H}} \quad (3)$$

$$\nabla \times \underline{\vec{H}} = (\sigma + j\omega\varepsilon)\underline{\vec{E}} \quad (4)$$

Ecuatia de ordinul II in E (se elimina H)

$$\nabla \times (\nabla \times \underline{\vec{E}}) = -j\,\omega\,\nabla \times (\mu \underline{\vec{H}}) = -\,\mathrm{j}\omega\mu\,\nabla \times \underline{\vec{H}} = -j\,\omega\,\mu(\sigma + j\omega\varepsilon)\underline{\vec{E}}$$

Mat.omogen

$$\nabla \times (\nabla \times \underline{\vec{E}}) = \nabla(\nabla \cdot \underline{\vec{E}}) - \Delta \underline{\vec{E}} = -\Delta \underline{\vec{E}}$$

(1)

$$\underline{\gamma}^2 \overset{\text{not}}{=} j\omega\mu(\sigma + j\omega\varepsilon)$$

$\underline{\gamma}$ constanta de propagare complexa.

$$\Delta \underline{\vec{E}} - j\,\omega\,\mu(\sigma + j\omega\varepsilon)\underline{\vec{E}} = 0$$

$$\Delta \underline{\vec{E}} - \underline{\gamma}^2\,\underline{\vec{E}} = 0$$

Ec.Helmholtz vectoriala complexa (ec. vectoriala complexa a undelor)

Ecuatia de ordinul II in H (se elimina E)

$$\nabla \times (\nabla \times \underline{\vec{H}}) = \nabla \times ((\sigma + j\omega\varepsilon)\underline{\vec{E}}) = (\sigma + j\,\omega\,\varepsilon)\,\nabla \times \underline{\vec{E}} = -j\,\omega\,\mu\,(\sigma + j\omega\varepsilon)\underline{\vec{H}}$$

$$\nabla \times (\nabla \times \underline{\vec{H}}) = \nabla(\nabla \cdot \underline{\vec{H}}) - \Delta \underline{\vec{H}} = -\Delta \underline{\vec{H}}$$

$$\Delta \underline{\vec{H}} - j\,\omega\,\mu(\sigma + j\omega\varepsilon)\underline{\vec{H}} = 0$$

$$\Delta \underline{\vec{H}} - \underline{\gamma}^2\,\vec{H} = 0$$

Ecuatiile in E si H sunt similare.

5.1. Propagarea undelor EM in dielectrici cu pierderi

Ecuatii de ordinul I

$$\nabla \cdot \underline{\vec{E}} = 0 \quad (1)$$

$$\nabla \cdot \underline{\vec{H}} = 0 \quad (2)$$

$$\nabla \times \underline{\vec{E}} = -j\omega\mu\underline{\vec{H}} \quad (3)$$

$$\nabla \times \underline{\vec{H}} = (\sigma + j\omega\varepsilon)\underline{\vec{E}} \quad (4)$$

$\underline{\vec{E}}$ si $\underline{\vec{H}}$ satisfac ecuatii identice de ordinul II, dar ele nu sunt independente, ci legate prin relatii de ordinul I.
*In rezolvare, se alege una din ecuatiile de ordinul II care se rezolva (de exemplu (**E)) iar cealalta marime se determina dintr-o ecuatie de ordinul I (de exemplu (3)).*

Ecuatii de ordinul II – in E sau H (ambele sunt de tip Helmholtz complex)

$$\Delta \underline{\vec{E}} - j\,\omega\,\mu(\sigma + j\omega\varepsilon)\underline{\vec{E}} = 0$$

$$\Delta \underline{\vec{E}} - \underline{\gamma}^2\,\underline{\vec{E}} = 0 \quad (**E)$$

$$\Delta \underline{\vec{H}} - j\,\omega\,\mu(\sigma + j\omega\varepsilon)\underline{\vec{H}} = 0$$

$$\Delta \underline{\vec{H}} - \underline{\gamma}^2\,\vec{H} = 0 \quad (**H)$$

$\underline{\gamma}$ constanta de propagare complexa.

$$\underline{\gamma} = \sqrt{j\,\omega\,\mu\,(\sigma + j\,\omega\,\varepsilon)}$$

$$\underline{\gamma} = \alpha + j\,\beta$$

$\alpha = \text{Real}\,(\gamma) > 0$ factor de atenuare [Np/m]

$\beta = \text{Imag}\,(\gamma) > 0$ constanta de faza (numar de unda) [rad/m]

5.1. Propagarea undelor EM in dielectrici cu pierderi

$\underline{\gamma}$ constanta de propagare complexa.

$$\underline{\gamma} = \sqrt{j\,\omega\,\mu\,(\sigma + j\,\omega\,\varepsilon)}$$

$$\underline{\gamma} = \alpha + j\,\beta$$

$\alpha = \text{Real}\,(\gamma) > 0$ factor de atenuare [Np/m]

$\beta = \text{Imag}\,(\gamma) > 0$ constanta de faza (numar de unda) [rad/m]

α si β se pot calcula in functie de parametrii de material si frecventa:

$$\alpha = \omega \sqrt{\frac{\mu\,\varepsilon}{2}\left[-1 + \sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]}$$

$$\beta = \omega \sqrt{\frac{\mu\,\varepsilon}{2}\left[+1 + \sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]}$$

α si β depinde de frecventa (prin ω explicit dar si proprietatile de material pot depinde de frecventa – materialele se numesc in acest caz "dispersive")

OBS:

daca $\sigma = 0$

atunci $\alpha = 0$

Similar cu ~

Obs: la linii in r.a.p. constanta de propagare complexa era

$$\underline{\gamma} = \sqrt{(R_l + j\,\omega\,L_l)\,(G_l + j\,\omega\,C_l)}$$

$L_l \sim \mu$

$G_l \sim \sigma$

$C_l \sim \varepsilon$

$R_l \sim$ Nu are correspondent, la linii e legat de conductorul care ghideaza

α si β se pot calcula in functie de parametrii lineici si de frecventa:

$$\alpha = \sqrt{\frac{1}{2}\left[R_l\,G_l - \omega^2 L_l C_l + \sqrt{(R_l\,G_l - \omega^2 L_l C_l)^2 + \omega^2 (L_l G_l + R_l C_l)^2}\right]}$$

$$\beta = \sqrt{\frac{1}{2}\left[-R_l\,G_l + \omega^2 L_l C_l + \sqrt{(R_l\,G_l - \omega^2 L_l C_l)^2 + \omega^2 (L_l G_l + R_l C_l)^2}\right]}$$

OBS: daca $R_l = 0$ si $G_l = 0$ ➡ atunci $\alpha = 0$

Intr-un mediu fara pierderi, undele se propaga fara atenuare.

5.2. Unda plana

- **Unda plana** = unda care depinde de o singura coordonata spatiala (de exemplu z) si de timp.

Obs: 1) Coordonata spatiala este cea de-a lungul careia are loc propagarea; 2) Intr-un plan perpendicular pe coordonata spatiala de care depinde, unda are aceeasi valoare la un moment de timp fixat.)

Undele plane nu exista in realitate (ar trebui generate de plane infinit extinse parcurse de curenti). Ele sunt importante pentru ca sunt simple si pot fi folosite ca aproximatii ale undelor reale la distante sufficient de departate de sursele care le genereaza - antene.

$$\vec{E}(x,y,z,t) = \vec{E}_x(x,y,z,t)\vec{\imath} + \vec{E}_y(x,y,z,t)\vec{\jmath} + \vec{E}_z(x,y,z,t)\vec{k}$$

Unda plana: $\vec{E}(z,t) = \vec{E}_x(z,t)\vec{\imath} + \vec{E}_y(z,t)\vec{\jmath} + \vec{E}_z(z,t)\vec{k}$

Unda plana in r.a.p – reprezentare in complex: $\underline{\vec{E}}(z) = \underline{E}_x(z)\vec{\imath} + \underline{E}_y(z)\vec{\jmath} + \underline{E}_z(z)\vec{k}$

Sa presupunem ca unda are doar componenta dupa directia Ox: $\underline{\vec{E}}(z) = \underline{E}_x(z)\vec{\imath}$

- **Polarizarea unei unde EM** = are legatura cu directia campului electric (detalii in sectiunea 5.5.)

Unda polarizata liniar dupa Ox.

Vom rezolva ecuatia de ordinul II in $\vec{E}$, in r.a.p, pentru o unda plana, polarizata dupa OX =>

5.2. Unda plana

Ecuatiile si solutiile sunt similare cu cele de la linii, cu E si H in loc de u si i.

Unda plana polarizata dupa Ox:

$$\Delta\vec{\underline{E}} - \underline{\gamma}^2\vec{\underline{E}} = 0$$

$$\vec{\underline{E}}(z) = \underline{E}_x(z)\vec{\imath}$$

$$\Delta\vec{\underline{E}} = \Delta\,\underline{E}_x(z)\vec{\imath} = \left(\frac{\partial^2\underline{E}_x(z)}{\partial x^2} + \frac{\partial^2\underline{E}_x(z)}{\partial y^2} + \frac{\partial^2\underline{E}_x(z)}{\partial z^2}\right)\vec{\imath} = \frac{\mathrm{d}^2\underline{E}_x(z)}{\mathrm{d}^2 z}\vec{\imath}$$

$$\frac{\mathrm{d}^2\underline{E}_x(z)}{\mathrm{d}z^2}\vec{\imath} - \underline{\gamma}^2\underline{E}_x(z)\vec{\imath} = 0 \quad \Rightarrow \quad \boxed{\frac{\mathrm{d}^2\underline{E}_x(z)}{\mathrm{d}z^2} - \underline{\gamma}^2\underline{E}_x(z) = 0}$$

Ec.Helmholtz scalara complexa

$$\underline{E}_x(z) = \underline{C}_1 e^{-\underline{\gamma}z} + \underline{C}_2 e^{+\underline{\gamma}z}$$

$$\underline{E}_x(z) = \underline{E}_d(z) + \underline{E}_i(z)$$

$\underline{C}_1$ si $\underline{C}_2$ sunt constante de integrare care se determina din datele problemei

La linii am rezolvat ecuatia de gradul II

$$\boxed{\frac{\mathrm{d}^2\underline{U}(z)}{\mathrm{d}z^2} = \underline{\gamma}^2\underline{U}(z)}$$

unda elementara directa de tensiune = $\underline{U}_d(z)$

$$\underline{U}(z) = \underline{C}_1 e^{-\underline{\gamma}z} + \underline{C}_2 e^{+\underline{\gamma}z}$$

$$\underline{U}(z) = \underline{U}_d(z) + \underline{U}_i(z)$$

unda elementara inversa de tensiune = $\underline{U}_i(z)$

Si campul magnetic se determina din rel. de ord.I

$$\nabla \times \vec{\underline{E}} = -j\omega\mu\vec{\underline{H}} \quad \Rightarrow \quad \vec{\underline{H}}(z) = \underline{H}_y(z)\vec{\jmath}$$

Unda plana este TEM (transveral electromagnetica)!

$$\underline{H}_y(z) = \frac{1}{\underline{Z}_c}\left(\underline{E}_d(z) - \underline{E}_i(z)\right)$$

$$\underline{Z}_c = \frac{\underline{E}_d}{\underline{H}_d} = -\frac{\underline{E}_i}{\underline{H}_i}$$

Raport de componente directe!

$$\underline{Z}_c = \frac{\underline{U}_d}{\underline{I}_d} = -\frac{\underline{U}_i}{\underline{I}_i}$$

Si curentul a fost determinat din rel. de ord.I

$$-\frac{\mathrm{d}\underline{U}(z)}{\mathrm{d}z} = (R_l + j\omega L_l)\,\underline{I}(z)$$

$$\underline{I}(z) = \frac{1}{\underline{Z}_c}\left(\underline{U}_d(z) - \underline{U}_i(z)\right)$$

$$\underline{Z}_c = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

Impedanța caracteristică complexă **a mediului**

Impedanța caracteristică complexă **a liniei**

$$\underline{Z}_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$$

5.3. Unda plana. Propagarea in medii nemarginite

$$\underline{E}_x(z) = \underline{C}_1 e^{-\underline{\gamma} z} + \underline{C}_2 e^{+\underline{\gamma} z}$$

$$\underline{E}_x(z) = \underline{E}_d(z) + \underline{E}_i(z)$$

In medii nemarginite $\underline{C}_2 = 0$

(nu exista unda inversa, altfel modulul campului ar tinde catre infinit pentru $z \to \infty$)

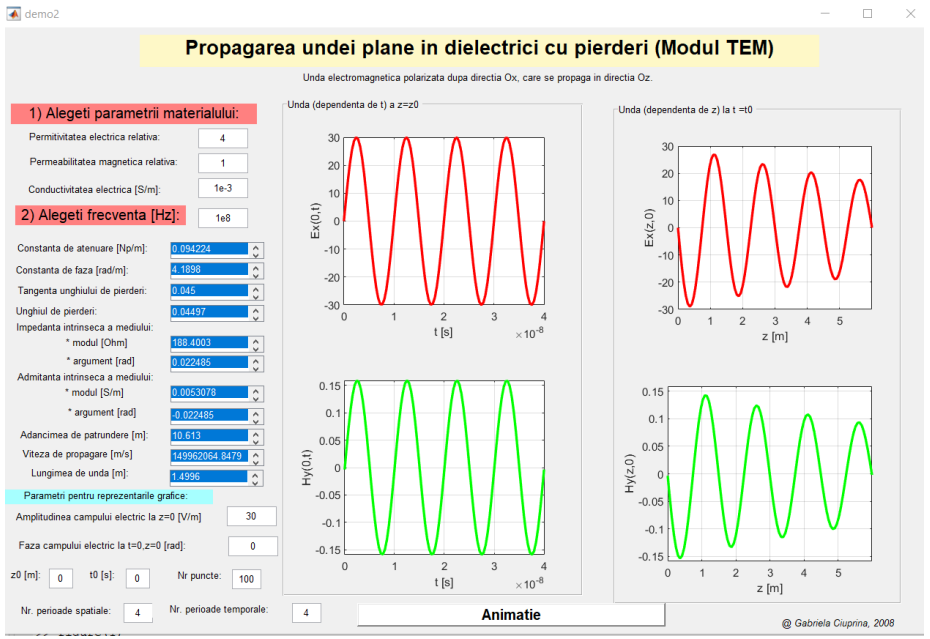
$$\underline{E}_x(z) = \underline{E}_d(z) = \underline{C}_1 e^{-\underline{\gamma} z}$$

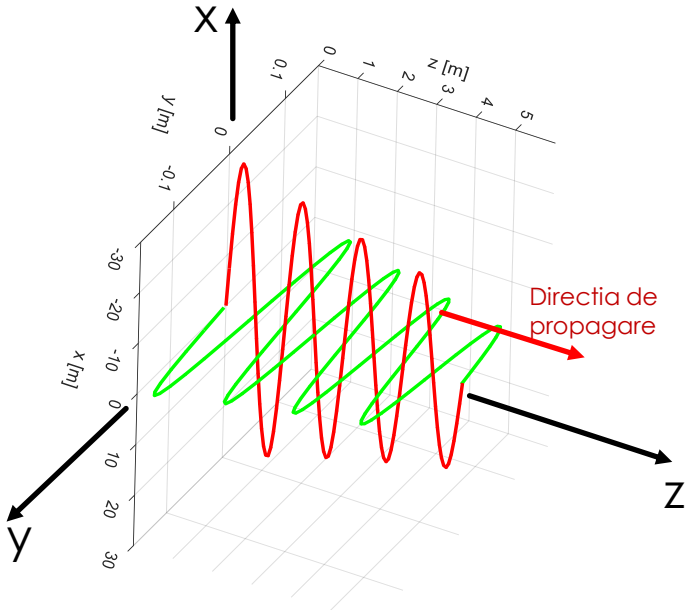
$$\underline{H}_y(z) = \underline{H}_d(z) = \frac{\underline{E}_d(z)}{\underline{Z}_c}$$

$$\underline{E}_x(0) = \underline{C}_1 = E_0 e^{j\varphi_0}$$

$$\underline{E}_x(z) = E_0 e^{j\varphi_0} e^{-\underline{\gamma} z} = E_0 e^{j\varphi_0} e^{-(\alpha + j\beta)z} = E_0 e^{-\alpha z} e^{j(-\beta z + \varphi_0)}$$

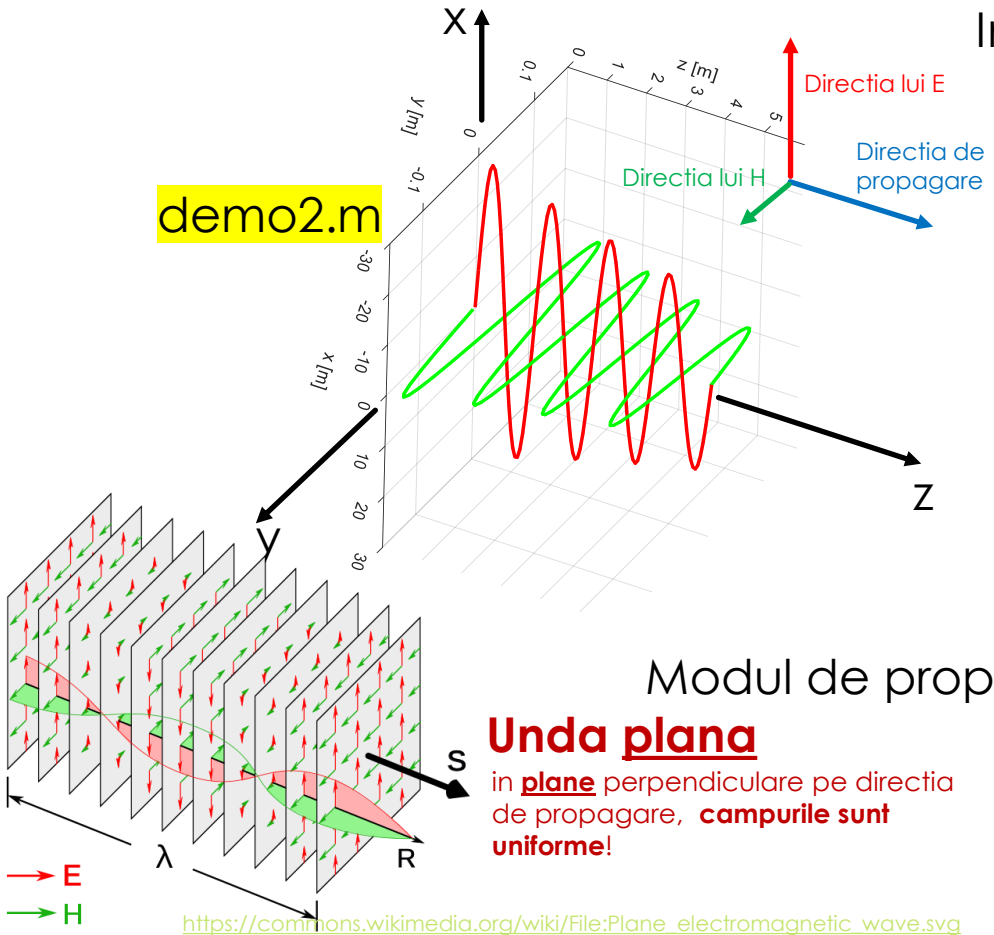
$$E_x(z,t) = E_0 e^{-\alpha z} \sqrt{2} \sin(\omega t - \beta z + \varphi_0)$$





Vedeti demo matlab – demo2.m

5.3. Unda plana. Propagarea in medii nemarginite



https://commons.wikimedia.org/wiki/File:Plane_electromagnetic_wave.svg

Unda plana

in **plane** perpendiculare pe directia de propagare, **campurile sunt uniforme**!

In general, pentru o unda plana

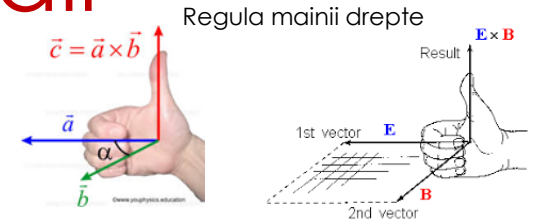
$$\overline{u}_E \times \overline{u}_H = \overline{u}_{\text{propagare}}$$

directia campului electric

directia campului magnetic

directia de propagare

Regula mainii drepte



La T. energiei am definit

$$\vec{E} \times \vec{H} = \vec{S}$$

Directia vectorului Poynting indica directia de propagare a undei.

(Detalii despre puteri in sectiunea 5.7.)

Modul de propagare a undei plane este **TEM – transversal electromagnetic**.

TEM

campul electric este perpendicular pe directia de propagare $\overline{u}_E \perp \overline{u}_{\text{propagare}}$

si

campul magnetic este perpendicular pe directia de propagare $\overline{u}_H \perp \overline{u}_{\text{propagare}}$

(Imaginati-va o unda TEM care nu este unda plana).

5.3. Unda plana. Propagarea in medii nemarginite

$$E_x(z,t) = E_d(z,t) = E_0 e^{-\alpha z}\sqrt{2}\sin(\omega t - \beta z + \varphi_0)$$

$$\omega T = 2\pi \qquad E_d(z,t) = E_d(z, t+T)$$

$$\beta\lambda = 2\pi \qquad E_d(z,t)e^{-\alpha\lambda} = E_d(z+\lambda, t)$$

	Spațiu	Timp
	λ [m] (lungime de unda)	T [s] (perioada temporala)
Frecvența unghiulara	$\beta = \frac{2\pi}{\lambda}\left[\frac{\text{rad}}{\text{m}}\right]$	$\omega = \frac{2\pi}{T}\left[\frac{\text{rad}}{\text{s}}\right]$

Punctele de faza constanta:

$$\omega t - \beta z + \varphi_{d0} = \text{constant}$$

$$\omega\, \mathrm{d}t - \beta\, \mathrm{d}z = 0$$

Viteza de propagare

$$\frac{dz}{dt} = \frac{\omega}{\beta} = v$$

$$v = \frac{\omega}{\beta} = \frac{2\pi}{T}\frac{\lambda}{2\pi} \Rightarrow v = \frac{\lambda}{T} \Rightarrow \qquad \lambda = vT$$

Relatiile sunt perfect similare cu cele de la linii.

5.3. Unda plana. Propagarea in medii nemarginite

$$E_x(z,t) = E_d(z,t) = E_0 e^{-\alpha z}\sqrt{2}\sin(\omega t - \beta z + \varphi_0)$$

α [Np/m] O masura a ratei spatiale de atenuare a undei.

$\frac{a}{b} = \ln\frac{a}{b}$ [Np]

$\frac{a}{b} = 1$[Np] => $a = e\ b$

Amplitudinea $A(z) = E_0 e^{-\alpha z}\sqrt{2}$

$A(0) = E_0\sqrt{2}$

$A(z) = A(0)\ e^{-\alpha z}$

$$-\alpha z = \ln\frac{A(z)}{A(0)} \rightarrow \boxed{\alpha = \frac{1}{z}\ln\frac{A(0)}{A(z)}}$$

O atenuare de 1 Nm/m inseamna o valoarea a amplitudinii unde la distanta de 1 m, de e ori mai mica decat cea de la $z = 0$.

Distanta δ la care amplitudinea undei se atenueaza de e ori se numeste **adancime de patrundere δ.**

Unda plana se atenueaza cand patrunde in medii cu pierderi, de exemplu in conductoare. Dupa 3-4 adancimi de patrundere unda devine neglijabila. In schimb in vid, unde $\sigma = 0, \alpha = 0$ si unda plana nu se atenueaza. Vedeti justificarea in slide-urile urmatoare.

$A(0) = E_0\sqrt{2}$

$A(\delta) = E_0 e^{-\alpha\delta}\sqrt{2}$

$A(\delta) = \frac{A(0)}{e}$

$e^{-\alpha\delta} = e^{-1}$

$$\boxed{\delta = \frac{1}{\alpha}}$$

5.3. Unda plana. Propagarea in medii nemarginite

$$\underline{E}_x(z) = \underline{E}_d(z) = \underline{C}_1 e^{-\underline{\gamma} z}$$

$$\vec{\underline{E}}(z) = \underline{E}_x(z)\vec{\imath}$$

$$\underline{H}_y(z) = \underline{H}_d(z) = \frac{\underline{E}_d(z)}{\underline{Z}_c}$$

$$\vec{\underline{H}}(z) = \underline{H}_y(z)\vec{\jmath}$$

$$\underline{Z}_c = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\underline{Y}_c = \frac{1}{\underline{Z}_c} = \sqrt{\frac{\sigma + j\omega\varepsilon}{j\,\omega\,\mu}}$$

Admitanta caracteristică

$$\left|\underline{H}_d(z)\right| = \frac{\left|\underline{E}_d(z)\right|}{\left|\underline{Z}_c\right|}$$

$$\underline{Z}_c = Z_c\, e^{j\,\theta}$$

$$\left|\underline{Z}_c\right| = Z_c \qquad \arg \underline{Z}_c = \theta$$

$$\arg \underline{H}_d(z) = \arg \underline{E}_d(z) - \arg \underline{Z}_c$$

$\sim$

$$\underline{U}(z) = \underline{U}_d(z) = \underline{C}_1 e^{-\underline{\gamma} z}$$

$$\underline{I}(z) = \frac{\underline{U}_d(z)}{\underline{Z}_c}$$

$$\underline{Z}_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$$

$$\underline{Y}_c = \frac{1}{\underline{Z}_c} = \sqrt{\frac{G_l + j\omega\, C_l}{R_l + \,j\,\omega\, L_l}}$$

Admitanta caracteristică

$$\left|\underline{I}_d(z)\right| = \frac{\left|\underline{U}_d(z)\right|}{\left|\underline{Z}_c\right|}$$

$$\arg I_d(z) = \arg \underline{U}_d(z) - \arg \underline{Z}_c$$

$$\underline{Z}_c^2 = \frac{j\omega\mu}{\sigma + j\omega\varepsilon} \Rightarrow Z_c^2 e^{j\,2\,\theta} = \frac{j\omega\mu\,(\sigma - j\,\omega\varepsilon)}{\sigma^2 + \omega^2\varepsilon^2} = \frac{\omega^2\mu\varepsilon + j\,\omega\mu\sigma}{\sigma^2 + \omega^2\varepsilon^2}$$

$$\Rightarrow 2\theta = \operatorname{atan}\frac{\omega\mu\sigma}{\omega^2\mu\varepsilon} \Rightarrow 2\theta = \operatorname{atan}\frac{\sigma}{\omega\,\varepsilon} \in \left[0, \frac{\pi}{2}\right] \Rightarrow \theta \in \left[0, \frac{\pi}{4}\right]$$

Campul magnetic e defazat in urma campului electric cu un unghi de cel mult de 45º, cauza acestui defazaj fiind impedanta mediului.

5.3. Unda plana. Propagarea in medii nemarginite.

Cazuri particulare:

a1) **In medii fara pierderi**: $\sigma = 0,\ \varepsilon = \varepsilon_0\varepsilon_r,\ \mu = \mu_0\mu_r$

$$\underline{\gamma} = \sqrt{(j\omega\mu)(\sigma + j\omega\varepsilon)} = \sqrt{-\omega^2\mu\varepsilon} = j\,\omega\,\sqrt{\mu\varepsilon}$$

$$\underline{\gamma} = \alpha + j\beta$$

$\alpha = 0$ Nu exista atenuare

$$\beta = \omega\,\sqrt{\mu\varepsilon}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon}}$$

Daca parametrii mediului nu depind de frecventa, atunci viteza de propagare nu depinde de frecventa. => Semnalele se propaga fara distorsiuni (toate armonicele se propaga cu aceeasi frecventa).

$$\underline{Z}_c = \sqrt{\frac{j\,\omega\mu}{\sigma + j\,\omega\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}} \in \mathrm{R}\,;\ \underline{Z}_c \overset{\text{not}}{=} Z_0 \qquad \theta = 0$$

Impedanta caracterstica este un numar real.

Campul magnetic e in faza cu campul electric.

~

a) Linia fara pierderi $(R_l = 0; \mathrm{G}_l = 0)$

$$\underline{\gamma} = \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)} = \sqrt{-\omega^2 L_l C_l} = j\,\omega\,\sqrt{L_l C_l}$$

$$\underline{\gamma} = \alpha + j\beta$$

$\alpha = 0$ Nu exista atenuare

$$\beta = \omega\,\sqrt{L_l C_l}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L_l C_l}}$$

Daca parametrii lineici nu depind de frecventa, atunci viteza de propagare nu depinde de frecventa. => Semnalele se propaga fara distorsiuni (toate armonicele se propaga cu aceeasi frecventa).

$$\underline{Z}_c = \sqrt{\frac{L_l}{C_l}} \in \mathrm{R}\,;\ \underline{Z}_c \overset{\text{not}}{=} Z_0$$

Impedanta caracterstica este un numar real.

5.3. Unda plana. Propagarea in medii nemarginite.

Cazuri particulare:

a2) **In vid**: $\sigma = 0,\ \varepsilon = \varepsilon_0,\ \mu = \mu_0$

$$c = \frac{1}{4\,\pi\,10^{-7}\cdot 8.854\cdot 10^{-12}} = 2.9979\ \cdot\ 10^8 \approx 3\ \cdot\ 10^8\ \text{m/s}$$

Viteza de propagare a luminii in vid.

Lumina este o unda electromagnetica.

In cazul anterior $\varepsilon_r = 1, \mu_r = 1$

$\alpha = 0$ Nu exista atenuare

$$\beta = \omega\sqrt{\mu_0\varepsilon_0}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \triangleq c$$

$$Z_0 = \sqrt{\frac{4\,\pi\,10^{-7}}{8.854\ \cdot\ 10^{-12}}} \approx 377\ \Omega$$

Impedanta vidului.

50 Ω

Impedanta tipica a cablurilor.

$$\underline{Z}_c = \sqrt{\frac{\mu_0}{\varepsilon_0}} \in \mathrm{R}\,;\ \underline{Z}_c \overset{\text{not}}{=} Z_0$$

Impedanta caracterstica este un numar real.

$\theta = 0$

Campul magnetic e in faza cu campul electric.

Obs: in dielectrici fara pierderi

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r\,\varepsilon_r}} < c$$

$\mu_r = 1$ (medii nemagnetice)
$\varepsilon_r > 1$ (medii dielectrice, nu vid)

$\frac{c}{v} \overset{\text{def}}{=} n = \sqrt{\mu_r\varepsilon_r}$ **Indicele de refactie al unui mediu.**

5.3. Unda plana. Propagarea in medii nemarginite.

Cazuri particulare:

b) **In conductoare foarte bune**: $\sigma \neq 0, \sigma \gg \omega\,\varepsilon$

$$\alpha = \omega\sqrt{\frac{\mu\,\varepsilon}{2}\left[-1+\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]} \approx \omega\sqrt{\frac{\mu\,\varepsilon}{2}\left[-1+\sqrt{\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]} \approx \omega\sqrt{\frac{\mu\,\varepsilon}{2}\left[-1+\frac{\sigma}{\omega\varepsilon}\right]} \approx \omega\sqrt{\frac{\mu\,\varepsilon}{2}\frac{\sigma}{\omega\varepsilon}} = \sqrt{\frac{\omega\mu\,\sigma}{2}}$$

$$\beta = \omega\sqrt{\frac{\mu\,\varepsilon}{2}\left[+1+\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]} \approx \sqrt{\frac{\omega\mu\,\sigma}{2}}$$

$$\underline{Z}_c = \sqrt{\frac{j\,\omega\mu}{\sigma + j\,\omega\varepsilon}} \qquad Z_c = |\underline{Z}_c| = \sqrt{\frac{\omega\mu}{\sqrt{\sigma^2+(\omega\varepsilon)^2}}} \approx \sqrt{\frac{\omega\mu}{\sigma}}$$

$$2\theta = \frac{\pi}{2} - \operatorname{atan}\frac{\omega\,\varepsilon}{\sigma} \approx \frac{\pi}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{4}$$

Campul magnetic e defazat in urma campului electric cu $\pi/4$.

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$v = \frac{\omega}{\beta} = \sqrt{\frac{2\,\omega}{\mu\,\sigma}}$$

Depinde de frecventa

Adancimea de patrundere in conductoare foarte bune:

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}}$$

5.3. Unda plana. Propagarea in medii nemarginite.

$\sigma_{Cu} = 5.8 \cdot 10^7$ S/m

f	50 Hz	1 MHz	10 GHz
δ	9 mm	66 µm	660 nm

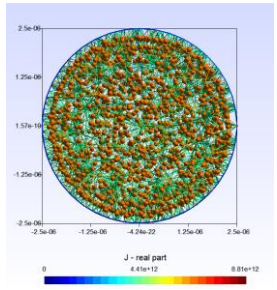
$$\delta = \frac{1}{\alpha} = \sqrt{\frac{1}{\pi f \mu \sigma}}$$

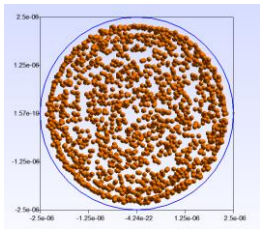
Efectul pelicular = fenomenul in care campul si curentul se afla intr-un strat subtire la suprafata conductorului.

Undele care patrund in corpurile conductoare se atenueaza foarte rapid (la adancimea maxima de 3-4 delta putem considera ca nu mai exista camp, adancimea de patrundere este cu atat mai mica cu cat frecventele sunt mai mari).

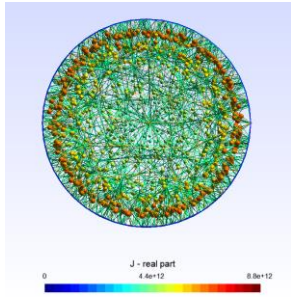
In aplicatii
- Ecranare EM – se realizeaza cu imprejmuiri conductoare avand grosimea egala cu cateva adancimi de patrundere.

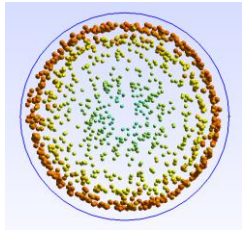
0.3 GHz



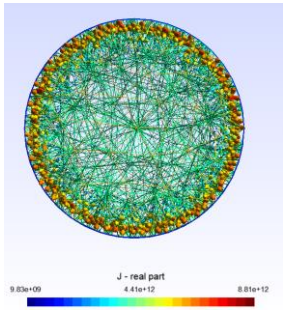


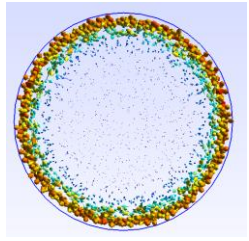
1 GHz

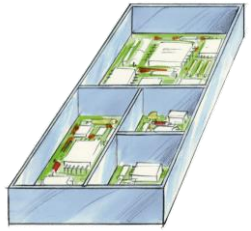




10 GHz







EMI-shielded fencing on the circuit board, which can be closed with a lid of conductive foam



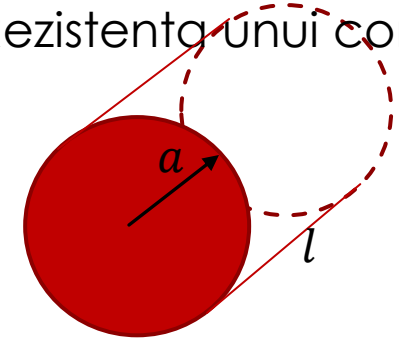
Mu-copper Faraday cages can be made to fit any type/shape/size of room.

https://hollandshielding.com/EMI-shielding-theory-applications

Model onelab disponibil Raza conductorului 2.5 um.

5.3. Unda plana. Propagarea in medii nemarginite.

Rezistenta unui conductor cilindric:

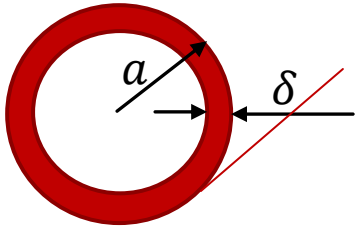


In c.c. – curentul se distribuie uniform in conductoare.

$$R = \frac{l}{\sigma \pi a^2}$$

$$R_{l,cc} = \frac{1}{\sigma \pi a^2} \quad \left[\frac{\Omega}{\mathrm{m}}\right]$$

Rezistenta lineica, de c.c., a unui conductor cilindric.



$$R = \frac{l}{\sigma 2\pi a\, \delta}$$

$$R_{l,ca-pp} = \frac{1}{\sigma 2\pi \delta a} > R_l, cc \quad \left[\frac{\Omega}{\mathrm{m}}\right]$$

Rezistenta lineica, de c.a., a unui conductor cilindric, in cazul unui effect pelicular pronuntat.

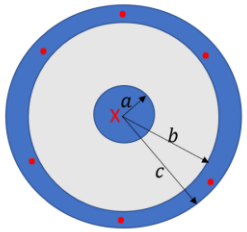
$$\frac{R_{l,ca-pp}}{R_{l,cc}} = \frac{a}{2\,\delta}$$

5.3. Unda plana. Propagarea in medii nemarginite.

Rezistenta unui cablu coaxial:

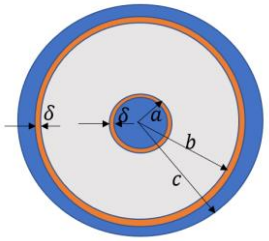
In c.c. – curentul se distribuie uniform in conductoare.

$$R = R_{\text{int}} + R_{\text{ext}} = \frac{l}{\sigma\pi a^2} + \frac{l}{\sigma(\pi c^2 - \pi b^2)} = \frac{l}{\sigma\pi}\left(\frac{1}{a^2} + \frac{1}{c^2 - b^2}\right)$$



$$R_{l,cc} = \frac{1}{\sigma\pi}\left(\frac{1}{a^2} + \frac{1}{c^2 - b^2}\right) \quad \left[\frac{\Omega}{\text{m}}\right]$$

Rezistenta lineica, de c.c., a unui cablu coaxial.



$$R = R_{\text{int}} + R_{\text{ext}} = \frac{l}{\sigma 2\pi a\,\delta} + \frac{l}{\sigma 2\pi b\,\delta} = \frac{l}{\sigma 2\pi\delta}\left(\frac{1}{a} + \frac{1}{b}\right)$$

$$R_{l,ca-pp} = \frac{1}{\sigma 2\pi\delta}\left(\frac{1}{a} + \frac{1}{b}\right) > R_l, cc \quad \left[\frac{\Omega}{\text{m}}\right]$$

Rezistenta lineica, de c.a., a unui cablu coaxial, in cazul unui effect pelicular pronuntat.

5.3. Unda plana. Propagarea in medii nemarginite.

Important!

Comportarea unui material in camp electromagnetic nu depinde numai de proprietatile lui electrice (ε), magnetice (μ) sau de conductie (σ) ci si de frecventa la care este folosit (ω).

$\operatorname{tg}\delta = \frac{\sigma}{\omega\varepsilon} \gg 1$ Materialul este un conductor foarte bun

$\operatorname{tg}\delta = \frac{\sigma}{\omega\varepsilon} \ll 1$ Materialul poate fi considerat fara pierderi.

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.

Daca mediul este marginit (de exemplu de un perete conductor), atunci poate aparea si unda inversa (care se intoarce in mediul in care exista si unda directa. Unda inversa apare dupa reflexia de perete. De asemenea, poate exista si o unda transmisa (refractata) in mediul vecin (de exemplu la trecerea din aer/vid intr-un mediu dielectric). Cand mediile au impedante caracteristice diferite apare, ca la linii, reflexia pe planul de interfata.

$$\Delta\underline{\vec{E}} - \underline{\gamma}^2\underline{\vec{E}} = 0$$

$$\frac{\mathrm{d}^2\underline{E}_x(z)}{\mathrm{d}z^2}\vec{\imath} - \underline{\gamma}^2\underline{E}_x(z)\vec{\imath} = 0$$

$$\frac{\mathrm{d}^2\underline{E}_x(z)}{\mathrm{d}z^2} - \underline{\gamma}^2\underline{E}_x(z) = 0$$

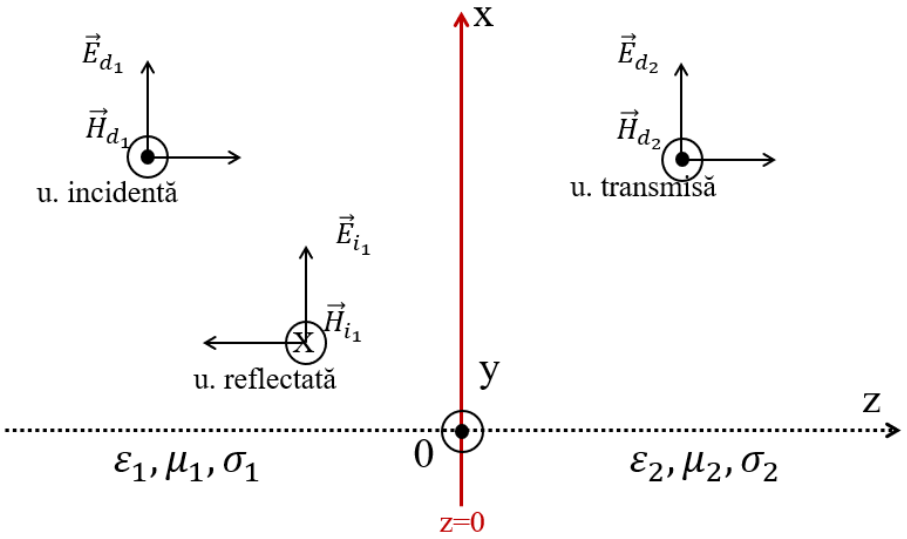
$$\underline{E}_x(z) = \underline{C}_1 e^{-\underline{\gamma}z} + \underline{C}_2 e^{+\underline{\gamma}z}$$

$$\underline{E}_x(z) = \underline{E}_d(z) + \underline{E}_i(z)$$

$$\underline{H}_y(z) = \frac{1}{\underline{Z}_c}\left(\underline{E}_d(z) - \underline{E}_i(z)\right)$$

$$\underline{Z}_c = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

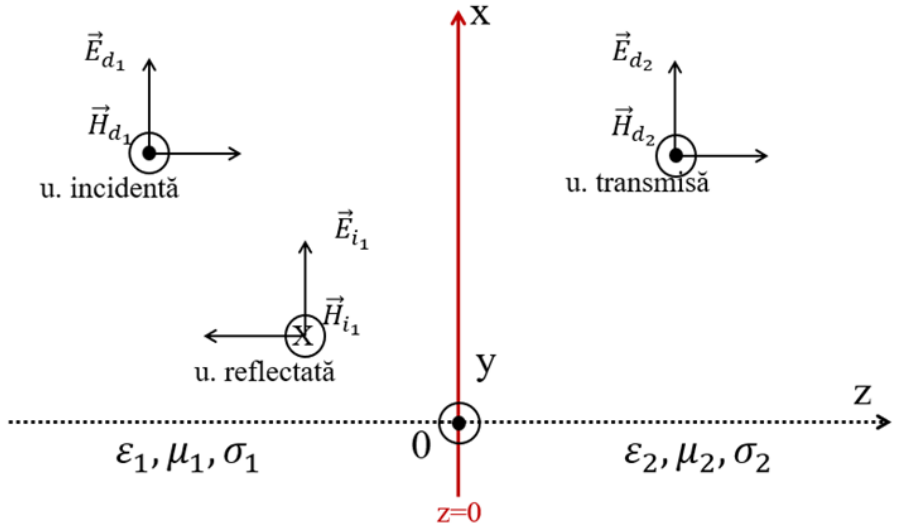
Incidenta normala (interfata dintre medii este perpendiculara pe directia de propagare) .



Pe interfata pp ca nu exista surse de camp (panze de current sau distributii superficiale de sarcina).

5.4. Unda plana. Reflexii si refractii.

Incidenta normala – rationament similar cu cel de la linii



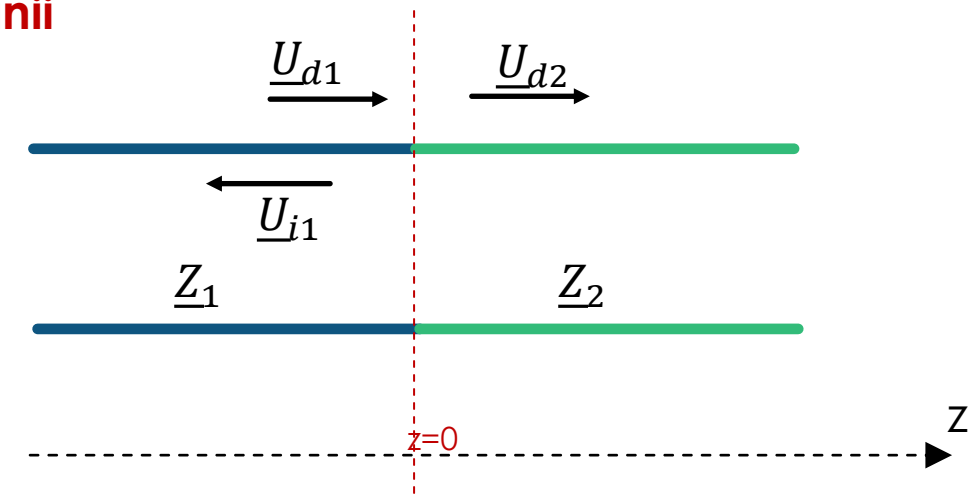
Pp, ca o unda plana se propaga in mediul 1 in directia axei Oz.

$$\underline{\vec{E}}_{\text{incident}}(z) = \underline{\vec{E}}_{d1}(z) = \underline{E}_{d1}(z)\vec{i}$$

$$\underline{\vec{H}}_{\text{incident}}(z) = \underline{\vec{H}}_{d1}(z) = \underline{H}_{d1}(z)\vec{j}$$

$$\underline{H}_{d1}(z) = \frac{\underline{E}_{d1}(z)}{\underline{Z}_1} \qquad \underline{Z}_1 = \sqrt{\frac{j\,\omega\,\mu_1}{\sigma_1 + j\,\omega\,\varepsilon_1}}$$

$\sim$



Pp, ca un semnal sinusoidal se propaga pe linia 1, in directia Oz.

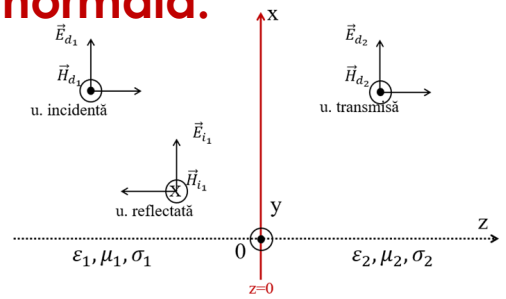
$$\underline{U}_{\text{incident}}(z) = \underline{U}_{d1}(z)$$

$$\underline{I}_{\text{incident}}(z) = \underline{I}_{d1}(z)$$

$$\underline{I}_{d1}(z) = \frac{\underline{U}_{d1}(z)}{\underline{Z}_1} \qquad \underline{Z}_1 = \sqrt{\frac{R_{l1} + j\,\omega\,L_{l1}}{G_{l1} + j\,\omega\,C_{l1}}}$$

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



Datorita schimbarii proprietatilor mediului, in mediul 1 exista si unda inversa (reflectata) iar in mediul 2 exista o unda transmisa (refractata).

$$\underline{\vec{E}}_{\text{reflectat}}(z) = \underline{\vec{E}}_{i1}(z) = \underline{E}_{i1}(z)\vec{i}$$

$$\underline{\vec{H}}_{\text{reflectat}}(z) = \underline{\vec{H}}_{i1}(z) = \underline{H}_{i1}(z)\vec{j}$$

$$\underline{H}_{i1}(z) = -\frac{\underline{E}_{i1}(z)}{\underline{Z}_1} \qquad \underline{Z}_1 = \sqrt{\frac{j\,\omega\,\mu_1}{\sigma_1 + j\,\omega\,\varepsilon_1}}$$

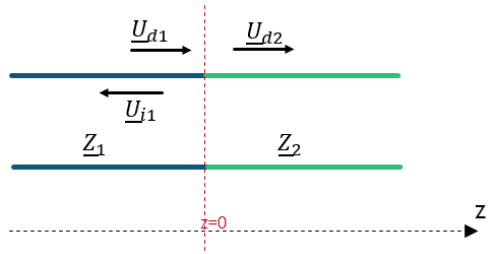
$$\underline{\vec{E}}_{\text{transmis}}(z) = \underline{\vec{E}}_{d2}(z) = \underline{E}_{d2}(z)\vec{i}$$

$$\underline{\vec{H}}_{\text{transmis}}(z) = \underline{\vec{H}}_{d2}(z) = \underline{H}_{d2}(z)\vec{j}$$

$$\underline{H}_{d2}(z) = \frac{\underline{E}_{d2}(z)}{\underline{Z}_2} \qquad \underline{Z}_2 = \sqrt{\frac{j\,\omega\,\mu_2}{\sigma_2 + j\,\omega\,\varepsilon_2}}$$

In mediul 2 nu exista unda inversa pentru ca el este pp. nemarginit (la dreapta).

~



Datorita schimbarii proprietatilor liniei, pe linia 1 exista si unda inversa (reflectata) iar pe linia 2 exista o unda transmisa (refractata).

$$\underline{U}_{\text{reflectat}}(z) = \underline{U}_{i1}(z)$$

$$\underline{I}_{\text{reflectat}}(z) = \underline{I}_{i1}(z)$$

$$\underline{I}_{i1}(z) = -\frac{\underline{U}_{i1}(z)}{\underline{Z}_1} \qquad \underline{Z}_1 = \sqrt{\frac{R_{l1} + j\,\omega\,L_{l1}}{G_{l1} + j\,\omega\,C_{l1}}}$$

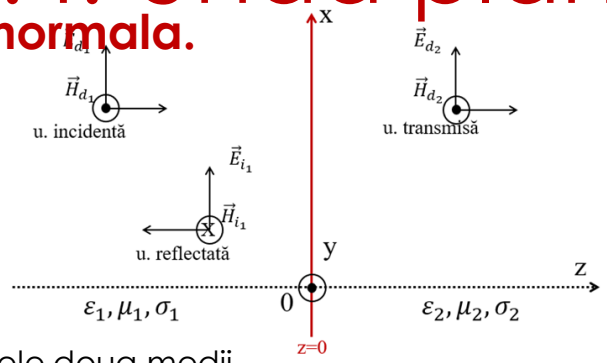
$$\underline{U}_{\text{transmis}}(z) = \underline{U}_{d2}(z)$$

$$\underline{I}_{\text{transmis}}(z) = \underline{I}_{d2}(z)$$

$$\underline{I}_{d2}(z) = \frac{\underline{U}_{d2}(z)}{\underline{Z}_2} \qquad \underline{Z}_2 = \sqrt{\frac{R_{l2} + j\,\omega\,L_{l2}}{G_{l2} + j\,\omega\,C_{l2}}}$$

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



Campul total in cele doua medii

$$\vec{\underline{E}}_1(z) = \underline{E}_{d1}(z)\vec{\imath} + \underline{E}_{i1}(z)\vec{\imath} \qquad \vec{\underline{E}}_2(z) = \underline{E}_{d2}(z)\vec{\imath}$$

$$\vec{\underline{H}}_1(z) = \underline{H}_{d1}(z)\vec{\jmath} + \underline{H}_{i1}(z)\vec{\jmath} \qquad \vec{\underline{H}}_2(z) = \underline{H}_{d2}(z)\vec{\jmath}$$

$$\underline{H}_{d1}(z) = \frac{\underline{E}_{d1}(z)}{\underline{Z}_1} \qquad \underline{H}_{d2}(z) = \frac{\underline{E}_{d2}(z)}{\underline{Z}_2}$$

$$\underline{H}_{i1}(z) = -\frac{\underline{E}_{i1}(z)}{\underline{Z}_1}$$

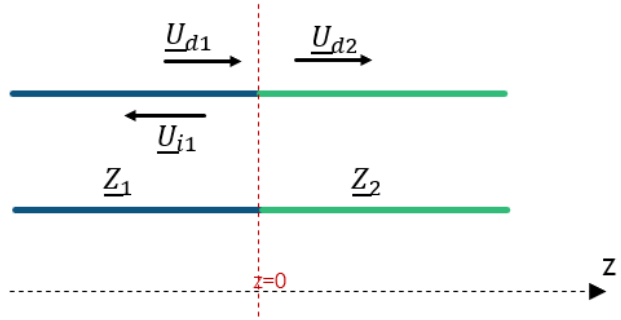
Corespund formelor locale ale legilor: LIE – se conserva componenta tangentiala a intensitatii cp.el, LCM fara panze de current – se conserva componenta tangentiala a intensitatii cp.mg. Aici campurile sunt tangentiale.

Conditii la interfata (z=0):

$$\underline{E}_1(0) = \underline{E}_2(0) \qquad \underline{E}_{d1}(0) + \underline{E}_{i1}(0) = \underline{E}_{d2}(0)$$

$$\underline{H}_1(0) = \underline{H}_2(0) \qquad \underline{H}_{d1}(0) + \underline{H}_{i1}(0) = \underline{H}_{d2}(0)$$

~



Tensiunea si curentul pe cele doua linii

$$\underline{U}_1(z) = \underline{U}_{d1}(z) + \underline{U}_{i1}(z) \qquad \underline{U}_2(z) = \underline{U}_{d2}(z)$$

$$\underline{I}_1(z) = \underline{I}_{d1}(z) + \underline{I}_{i1}(z) \qquad \underline{I}_2(z) = \underline{I}_{d2}(z)$$

$$\underline{I}_{d1}(z) = \frac{\underline{U}_{d1}(z)}{\underline{Z}_1} \qquad \underline{I}_{d2}(z) = \frac{\underline{U}_{d2}(z)}{\underline{Z}_2}$$

$$\underline{I}_{i1}(z) = -\frac{\underline{U}_{i1}(z)}{\underline{Z}_1}$$

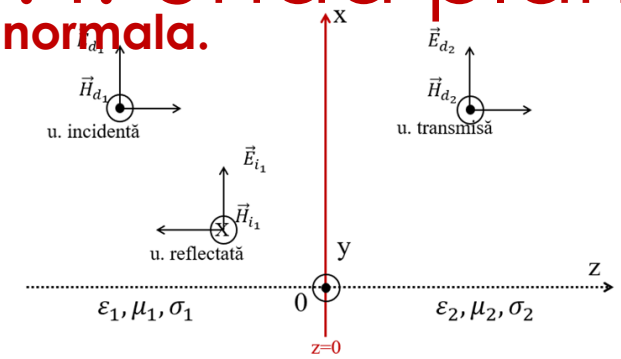
Conditii in punctul de conexiune al celor doua linii (z=0):

$$\underline{U}_1(0) = \underline{U}_2(0) \qquad \underline{U}_{d1}(0) + \underline{U}_{i1}(0) = \underline{U}_{d2}(0)$$

$$\underline{I}_1(0) = \underline{I}_2(0) \qquad \underline{I}_{d1}(0) + \underline{I}_{i1}(0) = \underline{I}_{d2}(0)$$

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



$$\underline{E}_{d1}(0) + \underline{E}_{i1}(0) = \underline{E}_{d2}(0)$$

$$\frac{\underline{E}_{d1}(0)}{\underline{Z}_1} - \frac{\underline{E}_{i1}(0)}{\underline{Z}_1} = \frac{\underline{E}_{d2}(0)}{\underline{Z}_2}$$

Pp. $\underline{E}_{d1}(0)$ cunoscut si calculam $\underline{E}_{i1}(0)$ si $\underline{E}_{d2}(0)$ in functie de el.

$$2\,\underline{E}_{d1}(0) = \underline{E}_{d2}(0)\left(1 + \frac{\underline{Z}_1}{\underline{Z}_2}\right)$$

$$\underline{E}_{d2}(0) = \frac{2\underline{Z}_2}{\underline{Z}_2 + \underline{Z}_1}\underline{E}_{d1}(0)$$

$$\underline{E}_{i1}(0) = \frac{\underline{Z}_2 - \underline{Z}_1}{\underline{Z}_2 + \underline{Z}_1}\underline{E}_{d1}(0)$$

$$\underline{E}_{d2}(0) = \underline{\tau}\,\underline{E}_{d1}(0)$$

$$\underline{E}_{i1}(0) = \underline{\Gamma}\,\underline{E}_{d1}(0)$$

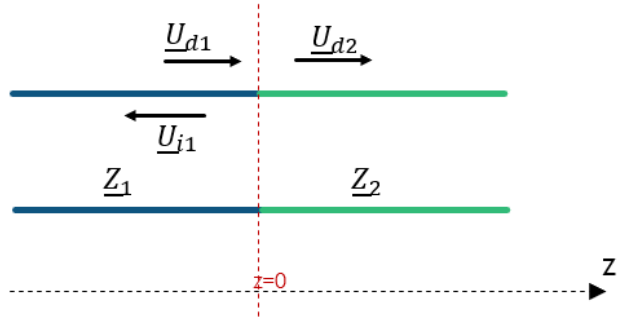
$$\underline{\Gamma} = \frac{\underline{Z}_2 - \underline{Z}_1}{\underline{Z}_2 + \underline{Z}_1}$$

Coeficient de reflexie

$$\underline{\tau} = \frac{2\,\underline{Z}_2}{\underline{Z}_2 + \underline{Z}_1}$$

Coeficient de transmisie

$$1 + \underline{\Gamma} = \underline{\tau} \qquad |\underline{\Gamma}| \leq 1$$



$$\underline{U}_{d1}(0) + \underline{U}_{i1}(0) = \underline{U}_{d2}(0)$$

$$\frac{\underline{U}_{d1}(0)}{\underline{Z}_1} - \frac{\underline{U}_{i1}(0)}{\underline{Z}_1} = \frac{\underline{U}_{d2}(0)}{\underline{Z}_2}$$

Pp. $\underline{U}_{d1}(0)$ cunoscut si calculam $\underline{U}_{i1}(0)$ si $\underline{U}_{d2}(0)$ in functie de el.

$$2\,\underline{U}_{d1}(0) = \underline{U}_{d2}(0)\left(1 + \frac{\underline{Z}_1}{\underline{Z}_2}\right)$$

$$\underline{U}_{d2}(0) = \frac{2\underline{Z}_2}{\underline{Z}_2 + \underline{Z}_1}\underline{U}_{d1}(0)$$

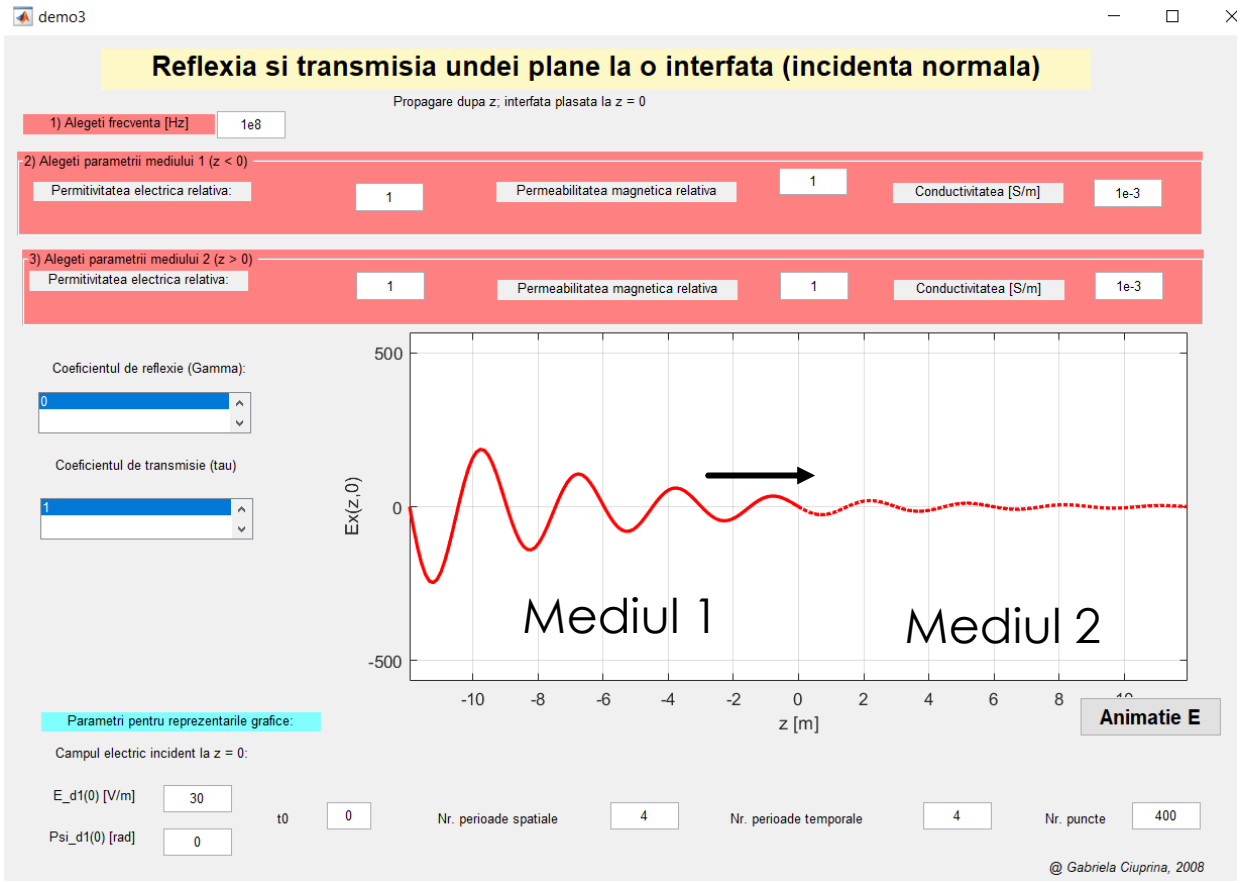
$$\underline{U}_{i1}(0) = \frac{\underline{Z}_2 - \underline{Z}_1}{\underline{Z}_2 + \underline{Z}_1}\underline{U}_{d1}(0)$$

$$\underline{U}_{d2}(0) = \underline{\tau}\,\underline{U}_{d1}(0)$$

$$\underline{U}_{i1}(0) = \underline{\Gamma}\,\underline{U}_{d1}(0)$$

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



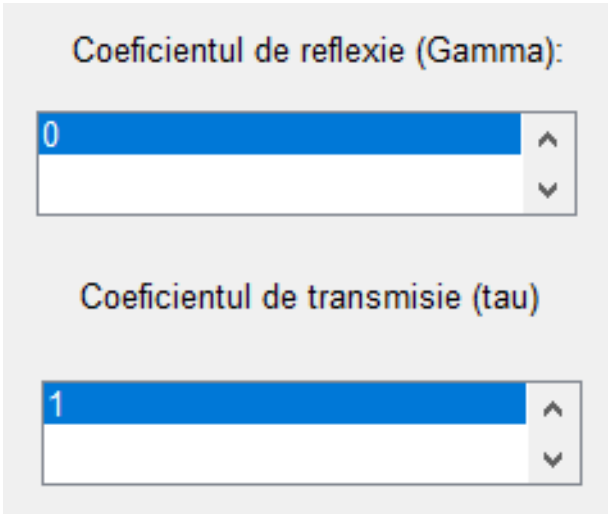
demo3.m

Mediul 1 (z<0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 10^{-3}$ S/m

Mediul 2 (z>0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_2 = 10^{-3}$ S/m

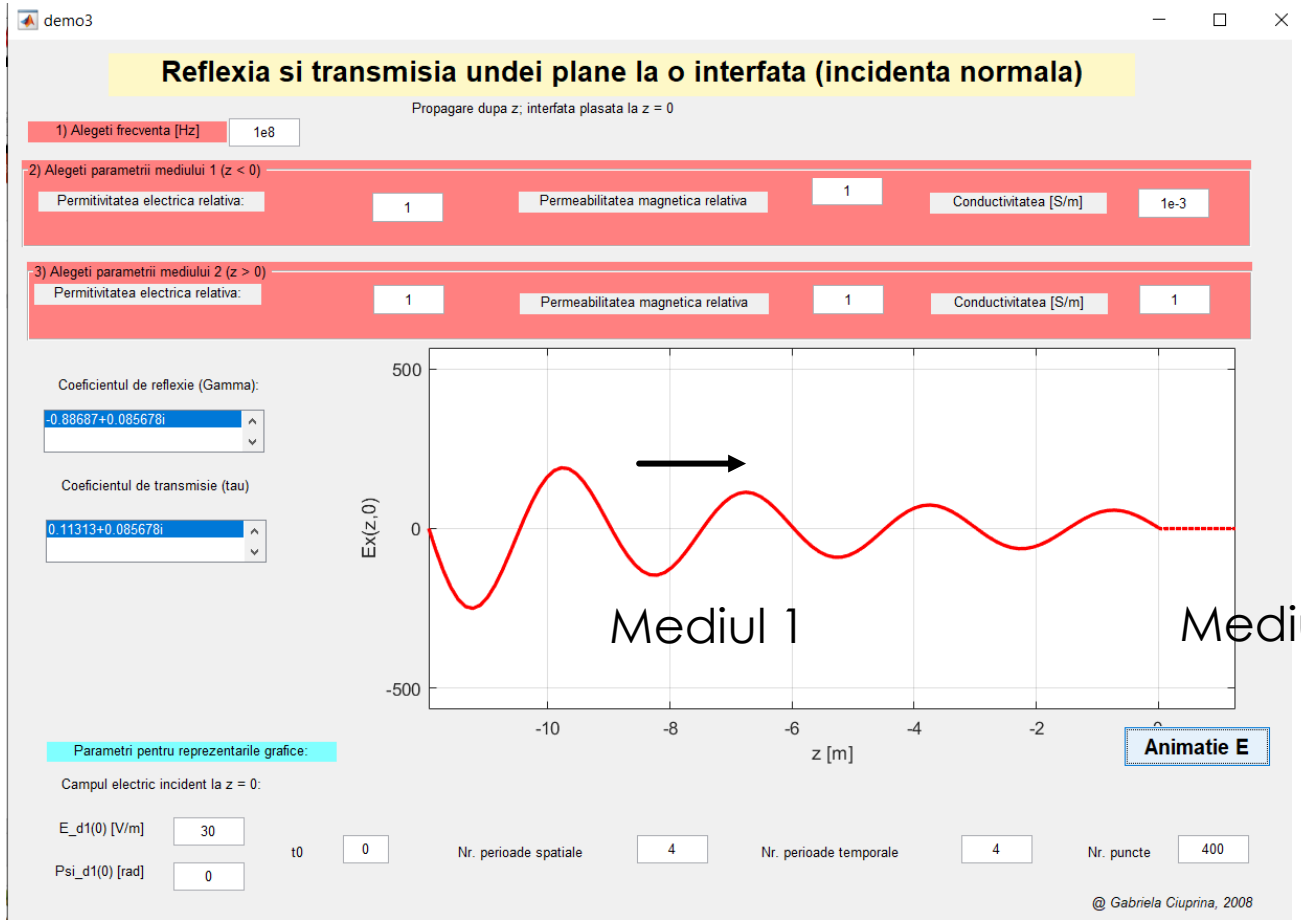


Nu exista reflexii

Totul se transmite

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



demo3.m

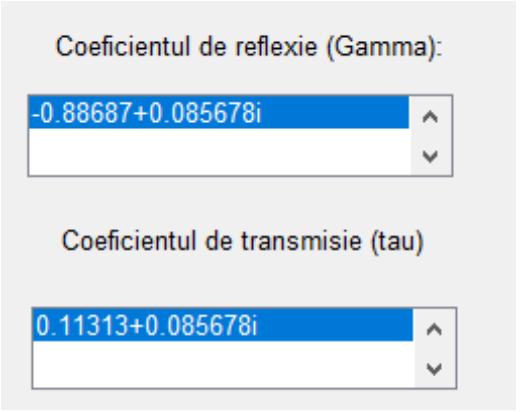
Mediul 1 (z<0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 10^{-3}$ S/m

Mediul 2 (z>0):

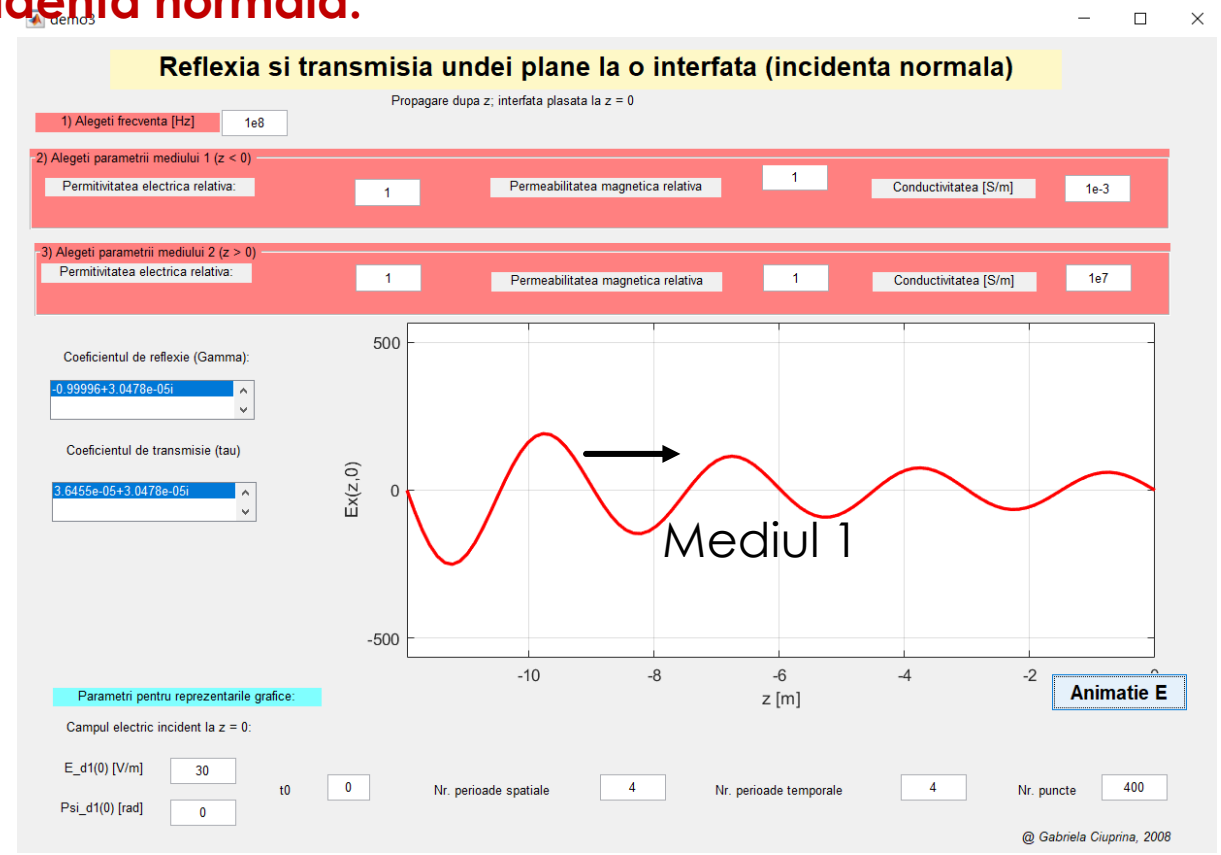
$\varepsilon_r = 1, \mu_r = 1, \sigma_2 = 1$ S/m

Unda patrunde foarte putin in mediul 2



5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



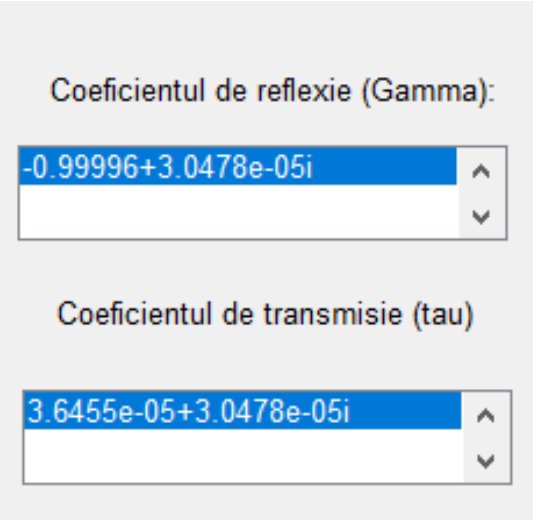
Mediul 1 (z<0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 10^{-3}$ S/m

Mediul 2 (z>0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_2 = 10^{7}$ S/m

Unda nu patrunde in mediul 2 (atat de putin incat nu este vizibil aici)



Ideal: $\sigma_2 \to \infty, \underline{Z}_2 = 0$

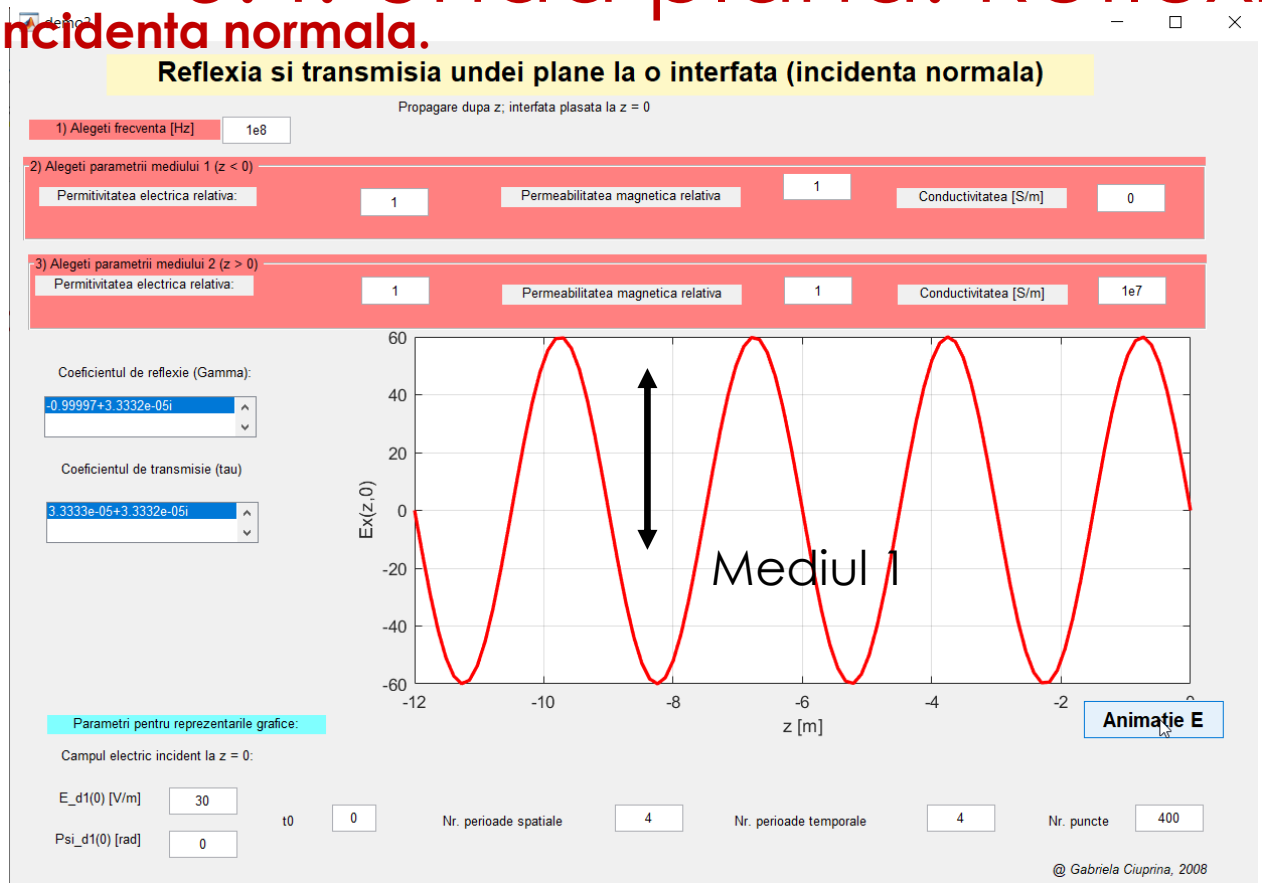
$\underline{\Gamma} = -1$ Totul se reflecta

$\underline{\tau} = 0$ Nimic nu se transmite

demo3.m

5.4. Unda plana. Reflexii si refractii.

Incidenta normala.



Mediul 1 (z<0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 0$ S/m

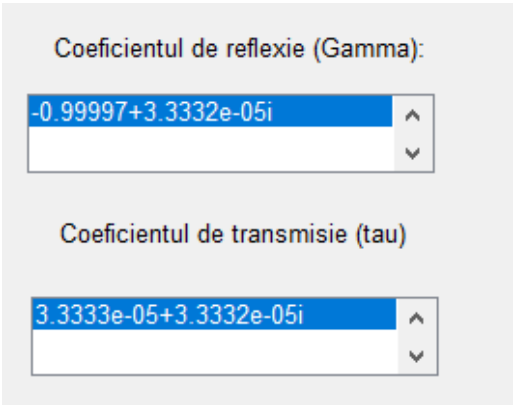
Nu exista atenuare

Mediul 2 (z>0):

$\varepsilon_r = 1, \mu_r = 1, \sigma_2 = 10^7$ S/m

Unda nu patrunde in mediul 2 (atat de putin incat nu este vizibil aici).

In mediul 1 apar **unde stationare (nu se propaga)**.



Ideal: $\sigma_2 \to \infty, \underline{Z}_2 = 0$

$\underline{\Gamma} = -1$ Totul se reflecta

$\underline{\tau} = 0$ Nimic nu se transmite

demo3.m

O unda sinusoidala neatenuata directa + unda inversa = o unda stationara

$$E_0 \sin(\omega t - \beta x) - E_0 \sin(\omega t + \beta x) = E_0 \left(\sin \frac{\omega t - \beta x + \omega t + \beta x}{2} \sin \frac{\omega t - \beta x - \omega t - \beta x}{2} \right)$$

$$= E_0 \sin(\omega t) \sin(\beta x)$$

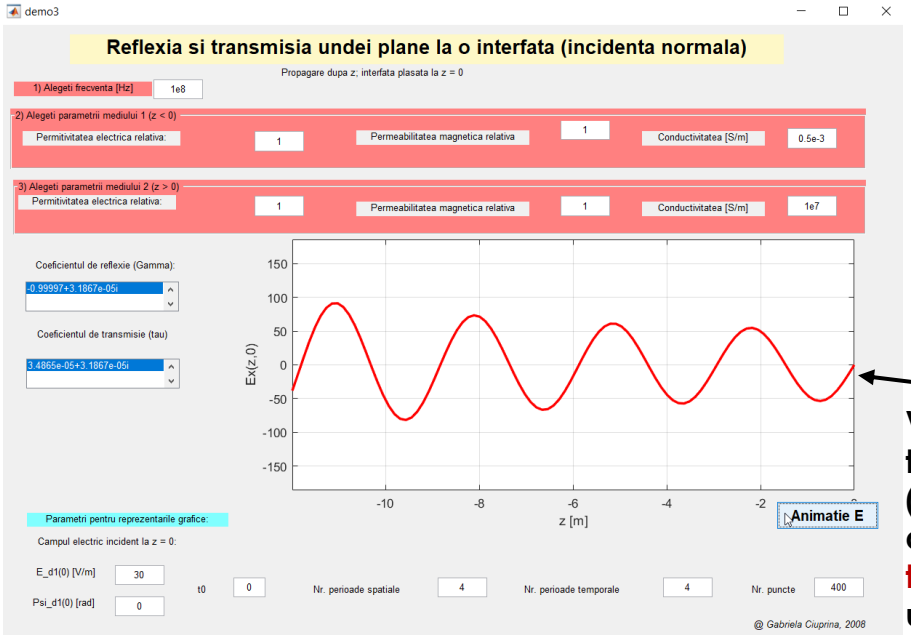
5.4. Unda plana. Reflexii si refractii.

Incidenta normala.

Obs 1: $\sigma_2 \to \infty$

inseamna perete (mediul 2) supraconductor (**PEC** = perfect electric conductor).

O conditie de interfata PEC este echivalenta cu impunerea componentei tangentiale a campului electric nula: $\overline{E}_t = 0$ sau $\overline{n} \times \overline{E} = 0$ (pentru ca $\sigma_2 \to \infty$ => $\overline{E}_2 = 0$ (altfel $\overline{J}_2 = \sigma_2 \overline{E}_2 \to \infty$ ceea ce nu are sens) => $\overline{E}_{t2} = 0 \Rightarrow \overline{E}_{t1} = 0$ => $\overline{E}_t = \overline{E}_{t1} = \overline{E}_{t2} = 0$).



demo3.m

Mediul 1 (z<0): $\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 0.5 \cdot 10^3$ S/m

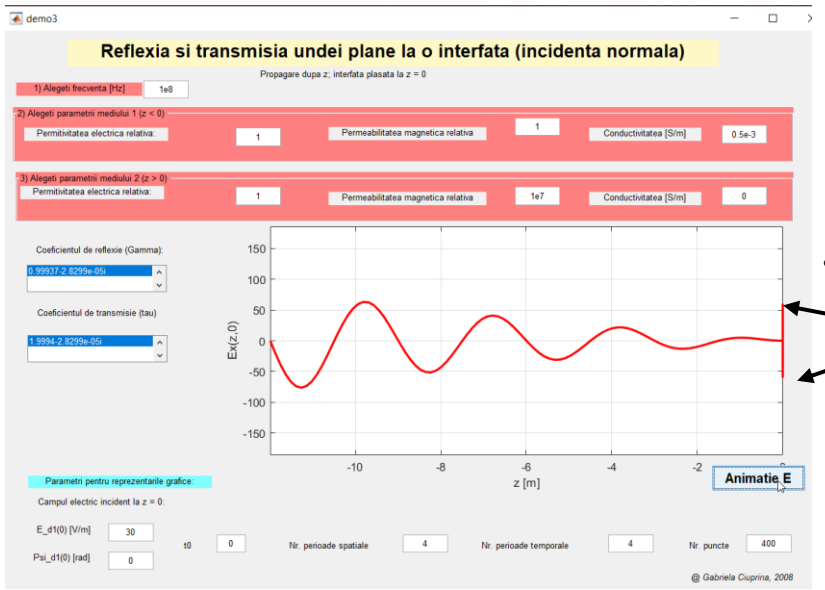
Mediul 2 (z>0): $\varepsilon_r = 1, \mu_r = 1, \sigma_2 = 10^7$ S/m

Valoarea aici este tot timpul zero (ca o coarda fixata la acest capat, sau ca tensiunea la capatul unei linii in scurt)

Obs 2: $\mu_2 \to \infty$

inseamna perete (mediul 2) perfect magnetic (PMC = perfect magnetic conductor).

O conditie de interfata PMC este echivalenta cu impunerea componentei tangentiale a campului magnetic nula: $\overline{H}_t = 0$ sau $\overline{n} \times \overline{H} = 0$ (pentru ca $\mu_2 \to \infty$ => $\overline{H}_2 = 0$ (altfel $\overline{B}_2 = \mu_2 \overline{H}_2 \to \infty$ ceea ce nu are sens) => $\overline{H}_{t2} = 0 \Rightarrow \overline{H}_{t1} = 0$ => $\overline{H}_t = \overline{H}_{t1} = \overline{H}_{t2} = 0$).



Mediul 1 (z<0): $\varepsilon_r = 1, \mu_r = 1, \sigma_1 = 0.5 \cdot 10^3$ S/m

Mediul 2 (z>0): $\varepsilon_r = 1, \mu_r = 10^7, \sigma_2 = 0$

Valoarea aici variaza intre o valoare minima si una maxima (ca o coarda lasata libera la acest capat, sau ca tensiunea la capatul unei linii in gol)

5.4. Unda plana. Reflexii si refractii.

demo3.m - Cod disponibil pe moodle

Ideea codului este simpla – din proprietatile mediului se calculeaza coeficientii de reflexie si de transmisie (linii 459 si 460), si apoi rezultatul adunarii undelor (linia 490). Se include totul intr-o bucla de timp si se deseneaza distributia spatiala, avand grija ca reprezentarile anterioare sa nu fie pastrate (linii 494 si 496). Animatia este o succesiune de reprezentari instantanee.

```
415   function zeta = impedanta_complexa(omega,mu,sigma,epsilon)
416 -   j = sqrt(-1);
417 -   zeta = sqrt(j*omega*mu/(sigma + j*omega*epsilon));
```

```
419   function [alpha,beta,lambda] = constanta_propagare(omega,mu,sigma,epsilon)
420 -   sigma_pe_omegaepsilon = sigma/(omega*epsilon);
421 -   muepsilon_pe_doi = mu*epsilon/2;
422 -   alpha = omega*...
423           sqrt(muepsilon_pe_doi*(sqrt(sigma_pe_omegaepsilon^2+1)-1)); %[Np/m]
424 -   beta = omega*...
425           sqrt(muepsilon_pe_doi*(sqrt(sigma_pe_omegaepsilon^2+1)+1)); %[rad/m]
426 -   lambda = 2*pi/beta;
```

Editor - D:\Gabriela\cursuri\Fizica3Electromagnetism_2022_2023\C5-undaPlana\matlab_demo\demo3.m

demo3.m

```
446 -   omega = 2*pi*f;
447
448 -   zeta1 = impedanta_complexa(omega,mu1*mu0,sig1,eps1*eps0);
449 -   zeta2 = impedanta_complexa(omega,mu2*mu0,sig2,eps2*eps0);
450 -   Gamma = (zeta2 - zeta1)/(zeta2 + zeta1);
451 -   tau = 2*zeta2/(zeta2 + zeta1);
452
453 -   set(handles.listbox_Gamma,'String',num2str(Gamma));
454 -   set(handles.listbox_tau,'String',num2str(tau));
455
456 -   [alpha1,beta1,lambda1] = constanta_propagare(omega,mu1*mu0,sig1,eps1*eps0);
457 -   [alpha2,beta2,lambda2] = constanta_propagare(omega,mu2*mu0,sig2,eps2*eps0);
458
459 -   coeficient_reflexie = (zeta2 - zeta1)/(zeta2 + zeta1);
460 -   coeficient_transmisie = 2*zeta2/(zeta2 + zeta1);
461
```

```
408
409   function [val_E] = camp_E_dir(Ex_0,alpha,omega,t,beta,z,psi0)
410 -   val_E = Ex_0*exp(-alpha*z).*sin(omega*t-beta*z+psi0);
411
412   function [val_E] = camp_E_inv(Ex_0,alpha,omega,t,beta,z,psi0)
413 -   val_E = Ex_0*exp(alpha*z).*sin(omega*t+beta*z+psi0);
```

Editor - D:\Gabriela\cursuri\Fizica3Electromagnetism_2022_2023\C5-undaPlana\matlab_demo\demo3.m

demo3.m

```
482 -   [val_E_mediul2] = ...
483           camp_E_dir(abs_Ed2_0_complex,alpha2,omega,t0,beta2,...
484           vector_z_mediul2,phase_Ed2_0_complex);
485 -   [val_E_mediul1_dir] = ...
486           camp_E_dir(Ed1_0,alpha1,omega,t0,beta1,...
487           vector_z_mediul1,Psid1_0);
488 -   [val_E_mediul1_inv] = camp_E_inv(abs_Ei1_0_complex,alpha1,omega,t0,beta1,...
489           vector_z_mediul1,phase_Ei1_0_complex);
490 -   val_E_mediul1 = val_E_mediul1_dir + val_E_mediul1_inv;
491
492 -   axes(handles.axes_E);
493 -   cla;
494 -   plot(vector_z_mediul2,val_E_mediul2,'r:','Linewidth',2);
495 -   hold on;
496 -   plot(vector_z_mediul1,val_E_mediul1,'r-','Linewidth',2);
497 -   axis([zmin zmax,-val_max_E ,val_max_E ]);
498 -   grid on;
499 -   xlabel('z [m]');
500 -   ylabel(strcat('Ex(z,',num2str(t0),')'));
```

5.5. Polarizarea undelor

- **Polarizarea unei unde plane** = se refera la modul de variatie in timp a directiei campului electric, mai precis la locul geometric al varfului vectorului $\vec{E}(z,t)$, vazut pe un plan perpendicular pe directia de propagare (front de unda).

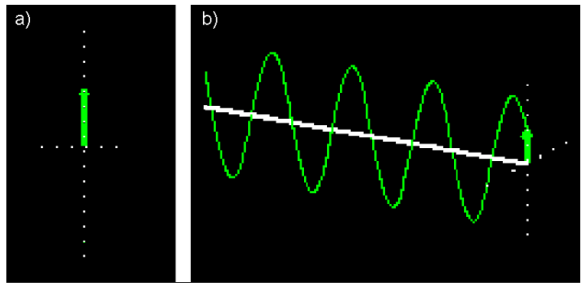
Pentru a putea comunica, doua antene trebuie sa produca unde polarizate similar.

Poate fi: **liniara**, **circulara**, **eliptica**

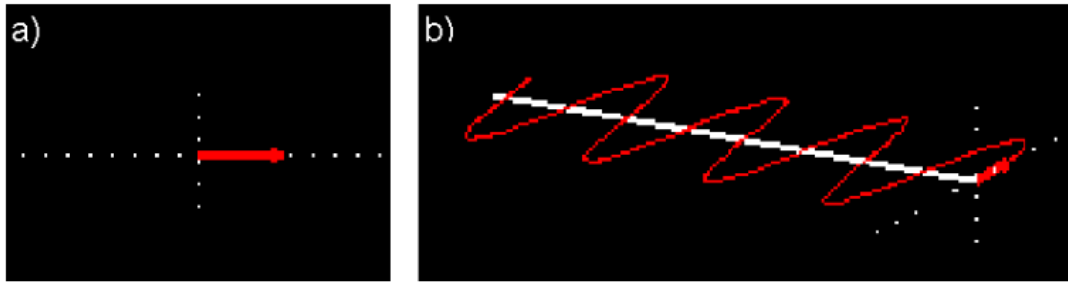
- varful vectorului $\vec{E}$ descrie (in timp) un segment/cerc/elipsa intr-un plan perpendicular pe directia de propagare (frontul undei).

radio AM (Vpol), radio FM (Vpol, Cpol, sau Epol), WiFi (Vpol), GPS (RHCP).

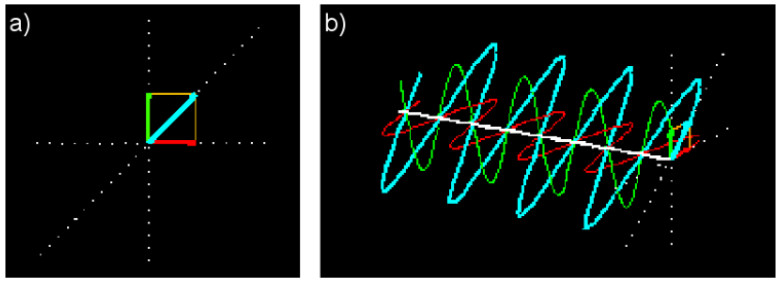
Polarizare liniara



Polarizare liniara - verticala
(folosita in radio AM si in WiFi)



Polarizare liniara - orizontala
(folosita la TV)



Polarizare liniara - oblica

Aici obtinuta din superpozitia a doua unde polarizate liniar (una verticala si una orizontala), de aceeasi frecventa, care sunt in faza si au aceeasi amplitudine (=> unghi de 45 de grade)

Vedeti animatii la https://shopdelta.eu/wave-polarisation_l2_aid893.html

5.5. Polarizarea undelor

$$\overline{E}(z,t) = E_1 \sin(\omega t - \beta z)\,\bar{i} + E_2 \sin(\omega t - \beta z + \Phi)\,\bar{j}$$

daca $\Phi = 0$ sau $\Phi = \pi$ **atunci**

$\overline{E}$ este polarizata liniar

altfel

daca $\Phi = \pm\frac{\pi}{2}$ si $E_1 = E_2$ **atunci**

$\overline{E}$ este polarizata circular

altfel

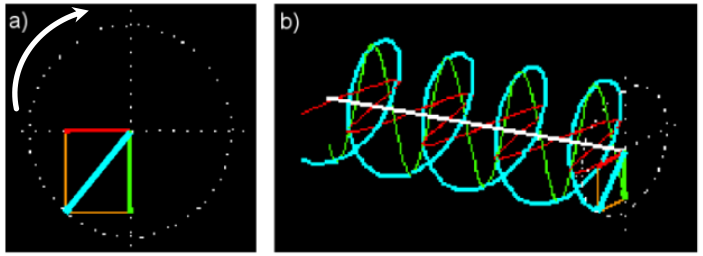
$\overline{E}$ este polarizata eliptic

*

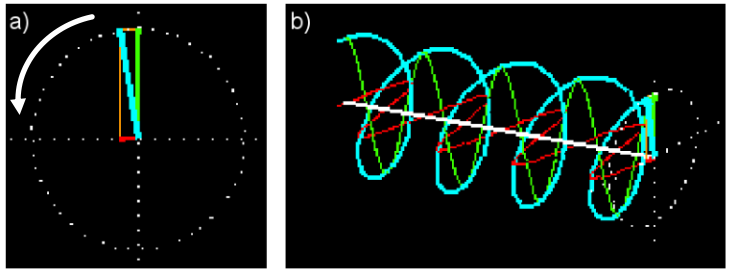
*

Daca cele doua unde sunt in faza, polarizarea este liniara, verticala pentru E2=0, orizontala cand E1=0 si oblica atunci cand E1>0, si E2>0, diferite. In cazul general, eliptic, defazajul este nenul, polarizarea devine circulara cand E1=E2 si defazajul este de 90 grade (cele doua unde polarizate liniar de amplitudine egala sunt in cuadratura atat spatial cat si temporal).

Polarizare circulara



Left handed circular polarized (LHCP)



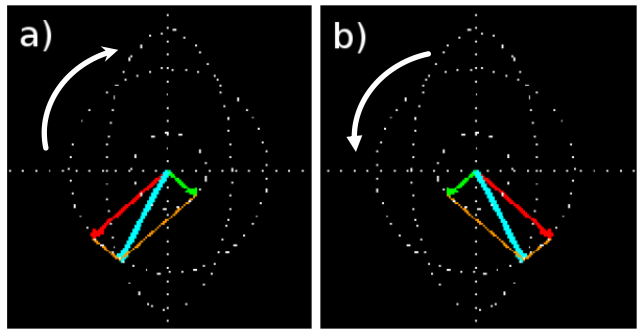
Right handed circular polarized (RHCP)

(folosite in GPS)

In figurile 2D - directia de propagare iese din foaie.
Luati mana dreapta astfel incat degetul mare deschis la 90 de grade (fata de celelalte degete intinse) sa indice directia de propagare (indreptat spre dvs). Indoiti celelalte 4 degete, varful lor indica directia de rotatie pentru cazul RHCP (trigonometric).

Polarizare eliptica

(folosita des in radio FM)



Left handed Right handed

Vedeti animatii la https://shopdelta.eu/wave-polarisation_l2_aid893.html

5.6. Reflexii si refractii. Incidenta oblica.

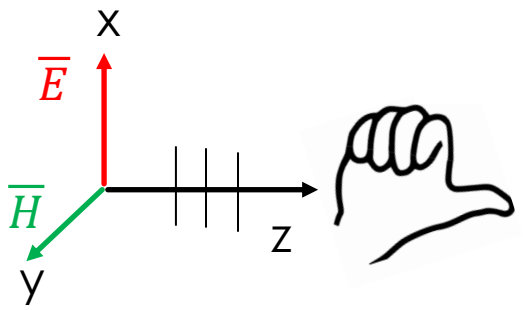
Evident ca undele nu se propaga numai normal catre interfeta dintre doua medii.
Pentru a analiza incidenta oblica, trebuie sa reformulam expresia undei plane in concordanta cu notatiile din literatura unde se foloseste o marime vectoriala notata cu $\overline{k}$, numita **vector de propagare**, orientat in directia si in sensul propagarii, care generealizeaza constanta de propagare $\underline{\gamma}$ pe care am folosit-o pana acum.

De aceea, vom renota versorii axelor de coordonate cu $\hat{x}, \hat{y}, \hat{z}$.

Reprezentarea in complex a unei plane directa polarizata liniar care se propaga dupa directia Oz se scrie ca:

$$\underline{\overline{E}}(z) = \underline{E_0}\, e^{-\underline{\gamma} z}\, \hat{x}$$

$$\underline{\overline{H}}(z) = \frac{\underline{E_0}}{\underline{Z_c}}\, e^{-\underline{\gamma} z}\, \hat{y}$$



$$\underline{\overline{E}}(\overline{r}) = \underline{E_0}\, e^{-\overline{k}\cdot\overline{r}}\, \hat{x}$$

$$\underline{\overline{H}}(\overline{r}) = \frac{\underline{E_0}}{\underline{Z_c}}\, e^{-\overline{k}\cdot\overline{r}}\, \hat{y}$$

Notam $\overline{k} = \underline{\gamma}\hat{z}$ vector de propagare

$\overline{r} = z\,\hat{z}$ vector de pozitie

$$\overline{k}\cdot\overline{r} = \underline{\gamma} z$$

Aceasta exprimare poate fi extensa.

5.6. Reflexii si refractii. Incidenta oblica.

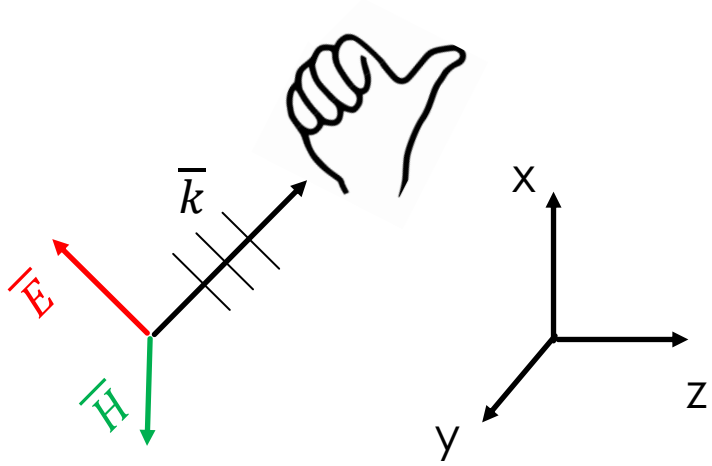
$\overline{k} = \underline{\gamma}\hat{k}$ **vector de propagare complex (sau numar de unda vectorial)**

$\hat{k}$ versorul directiei de propagare

$\overline{k} = k_x\hat{x} + k_y\hat{y} + k_z\hat{z}$

In cazul undei plane care se propaga dupa Oz, $k_x = 0, k_y = 0, k_z = \underline{\gamma}$

$\overline{r} = x\,\hat{x} + y\,\hat{y} + z\,\hat{z}$ vector de pozitie



Reprezentarea unei unde plane care se propaga intr-o directie $\hat{k}$:

$$\underline{\overline{E}}(\overline{r}) = \underline{\overline{E_0}}\, e^{-\overline{k}\cdot\overline{r}}$$

$$\underline{\overline{H}}(\overline{r}) = \underline{\overline{H_0}}\, e^{-\overline{k}\cdot\overline{r}}$$

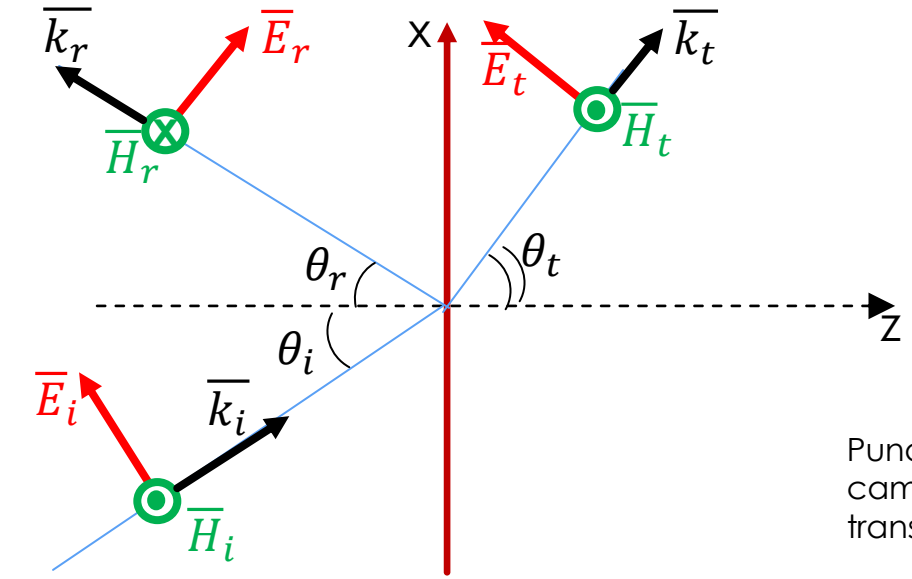
Mod TEM: $\overline{E} \perp \overline{H}, \overline{E} \perp \overline{k}, \overline{H} \perp \overline{k}$

$$\underline{\overline{H}} = \frac{1}{\underline{Z_c}}\hat{k} \times \underline{E}$$

$$\overline{E} = \underline{Z_c}\,\underline{\overline{H}} \times \hat{k}$$

5.6. Reflexii si refractii. Incidenta oblica.

Incidenta obliga se reduce la studiul urmatoarelor doua situatii (o unda plana cu orientare arbitrara este o superpozitie de astfel de unde una polarizata paralel si alta perpendicular pe planul de incidenta (determinat de directia de propagare si normala la planul interfetei).



Unda polarizata paralel cu planul de incidenta ($\overline{E}$ are doar componente in planul de incidenta).

Aplicatie – antenna parabolica
https://www.analyzemath.com/parabola/parabola_work.html

Planul xoz (determinat de vectorul de propagare al undei incidente si normala la suprafata de separatie – aici versorul axei Oz) = plan de incidenta.

θ_i = unghi de incidenta
θ_r = unghi de reflexie
θ_t = unghi de transmisie

Se demonstreaza ca (optica geometrica)

$$\theta_i = \theta_r$$

$$n_1 \sin\theta_i = n_2 \sin\theta_t \quad \text{(Snell)}$$

$n = \sqrt{\mu_r \varepsilon_r} = c/v$ Indice de refractie

Punand conditiile la interfata (continuitatea componentelor tangetiale ale campului electric si magnetic), se deduc relatiile dintre undele reflectate si transmise si cea incidenta:

$$\underline{\Gamma}_{||} = \underline{\Gamma}_H = \frac{\underline{E}_{r0}}{\underline{E}_{i0}} = \frac{\underline{Z}_2 \cos\theta_t - \underline{Z}_1 \cos\theta_i}{\underline{Z}_2 \cos\theta_t + \underline{Z}_1 \cos\theta_i}$$

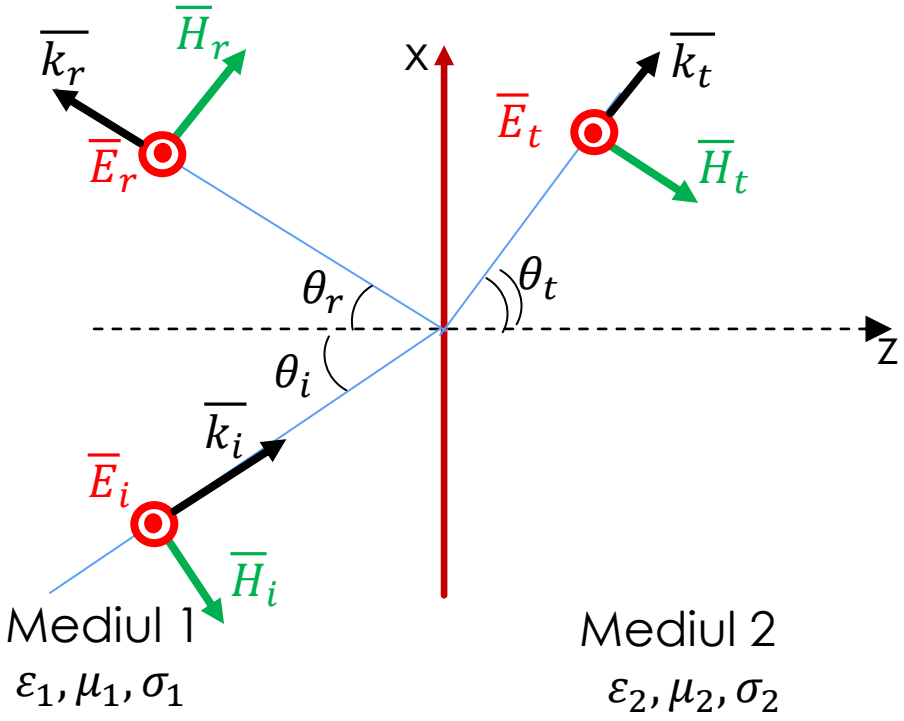
Coeficient de reflexie la incidenta oblica si polarizare paralela.

$$\underline{\tau}_{||} = \underline{\tau}_H = \frac{\underline{E}_{t0}}{\underline{E}_{i0}} = \frac{2\,\underline{Z}_2 \cos\theta_i}{\underline{Z}_2 \cos\theta_t + \underline{Z}_1 \cos\theta_i}$$

Coeficient de transmisie la incidenta oblica si polarizare paralela.

Formulele Fresnel pentru polarizare paralela.

5.6. Reflexii si refractii. Incidenta oblica.



Unda polarizata perpendicular pe planul de incidenta ($\overline{E}$ are doar componente perpendicular pe planul de incidenta).

Dupa reflectare, undele isi schimba polarizarea, RHCP devine LHCP, si GPS nu receptioneaza deci unde reflectate, ci doar cele venite direct de la satelit.

Punand conditiile la interfata, se deduc relatiile dintre undele reflectate si transmise si cea incidenta.

$$\underline{\Gamma}_{\perp} = \underline{\Gamma}_E = \frac{\underline{E}_{r0}}{\underline{E}_{i0}} = \frac{\underline{Z}_2 \cos\theta_i - \underline{Z}_1 \cos\theta_t}{\underline{Z}_2 \cos\theta_i + \underline{Z}_1 \cos\theta_t}$$

$$\underline{\tau}_{\perp} = \underline{\tau}_E = \frac{\underline{E}_{t0}}{\underline{E}_{i0}} = \frac{2\,\underline{Z}_2 \cos\theta_i}{\underline{Z}_2 \cos\theta_i + \underline{Z}_1 \cos\theta_t}$$

Coeficient de reflexie la incidenta oblica si polarizare perpendiculara.

Coeficient de transmisie la incidenta oblica si polarizare perpendiculara.

Formulele Fresnel pentru polarizare perpendiculara.

Utile in aplicatii practice (cablu optic):

Obs:

$$1 + \underline{\Gamma}_{||} = \frac{\cos\theta_t}{\cos\theta_i}\,\underline{\tau}_{||}$$

$$1 + \underline{\Gamma}_{\perp} = \underline{\tau}_{\perp}$$

Incidenta normala $\theta_i = 0$, de unde rezulta $\theta_r = 0, \theta_t = 0$.

- Unghiul critic (de incidenta pentru reflexie totala) – unghiul pentru care nu exista unda transmisa (30 grade pentru interfata sticla-aer).
- Unghiul Brewster (de incidenta) – unghiul pentru care nu exista unda reflectata (63.4 grade pentru interfata sticla-aer).

Vedeti si cod demonstrativ la
http://sweet.ua.pt/smota/EMSoftware/

pentru reprezentari grafice ale modulele acestor coeficienti, in functie de θ_i pentru interfetele aer/sticla si sticla/aer.

5.7. Puterea transmisa

Reamintire din capitolul de teoreme

[3.2] Teorema energiei electromagnetice

Teorema energiei electromagnetice (Poynting)

Teorema energiei electromagnetice - enunţ (medii liniare şi imobile)

Puterea transferată de câmpul electromagnetic, de la exteriorul la interiorul unui domeniu prin frontiera sa este egală cu suma dintre puterea transferată corpurilor conductoare din domeniu şi viteza de creştere a energiei electromagnetice din domeniu.

$$P_{\Sigma,\text{ext}\to\text{int}} = P_J + \frac{\mathrm{d}W_{\text{em}}}{\mathrm{d}t} \qquad [\mathrm{W}]$$

Teorema energiei
- este un bilant de puteri si energii în acord cu primul principiu al termodinamicii (global dar si local);
-evidentiaza capacitatea câmpului elmg. de a **transfera** energie (inclusiv în vid);
-evidentiaz a capacitatea câmpului elmg. de a **acumula** energie.

Forma locala:

$$-\operatorname{div}\left(\overline{E}\times\overline{H}\right) = \overline{J}\cdot\overline{E} + \frac{\partial w_e}{\partial t} + \frac{\partial w_m}{\partial t}$$

Conform legii transformării energiei în conductoare

$$p = \overline{J}\cdot\overline{E} \qquad \left[\frac{\mathrm{W}}{\mathrm{m}^3}\right]$$

Notăm $\overline{S}$ - vector Poynting

$$\overline{S} = \overline{E}\times\overline{H} \qquad \left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

şi $w_{\text{em}} = w_e + w_m$ - densitatea de energie a câmpului electromagnetic.

Forma locală a T.energiei în medii imobile şi liniare

$$-\operatorname{div}\overline{S} = p + \frac{\partial w_{\text{em}}}{\partial t}$$

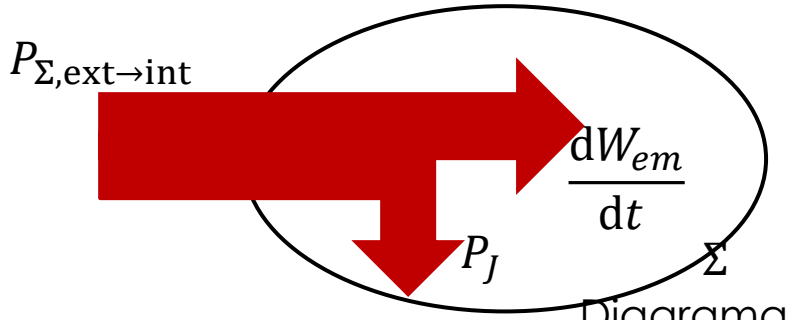


Diagrama Sankey https://en.wikipedia.org/wiki/Sankey_diagram

Forma globala:

$$\int_{\mathcal{D}_\Sigma} -\operatorname{div}\overline{S}\,\mathrm{dv} = \int_{\mathcal{D}_\Sigma} p\,\mathrm{dv} + \int_{\mathcal{D}_\Sigma} \frac{\partial w_{\text{em}}}{\partial t}\,\mathrm{dv}$$

$$\int_{\mathcal{D}_\Sigma} p\,\mathrm{dv} \overset{\text{not}}{=} P_J \qquad [\mathrm{W}]$$

$$\int_{\mathcal{D}_\Sigma} \frac{\partial w_{\text{em}}}{\partial t}\,\mathrm{dv} = \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{D}_\Sigma} w_{\text{em}}\,\mathrm{dv} \overset{\text{not}}{=} \frac{\mathrm{d}W_{\text{em}}}{\mathrm{d}t}$$

$$P_{\Sigma,\text{ext}\to\text{int}} = -\oint_\Sigma \overline{S}\cdot\overline{\mathrm{dA}}$$

$$P_{\Sigma,\text{ext}\to\text{int}} = P_J + \frac{\mathrm{d}W_{\text{em}}}{\mathrm{d}t} \qquad [\mathrm{W}]$$

5.7. Puterea transmisa

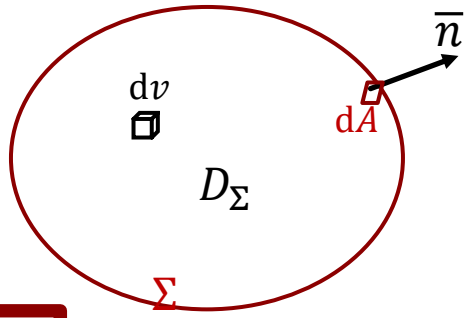
Forma teoremei in regim armonic permanent:

Forma locala:

$$-\operatorname{div}\left(\underline{\overline{E}}\times\underline{\overline{H}}^{*}\right)=\underline{\overline{E}}\cdot\underline{\overline{J}}^{*}+2\,j\,\omega\left(\frac{\underline{\overline{B}}\cdot\underline{\overline{H}}^{*}}{2}-\frac{\underline{\overline{E}}\cdot\underline{\overline{D}}^{*}}{2}\right)$$

Forma integrala:

$$-\oint_{\Sigma}\left(\underline{\overline{E}}\times\underline{\overline{H}}^{*}\right)\cdot\overline{n}\;\mathrm{d}A=\int_{D_\Sigma}\underline{\overline{E}}\cdot\underline{\overline{J}}^{*}\,\mathrm{d}v+2\,j\,\omega\int_{D_\Sigma}\left(\frac{\underline{\overline{B}}\cdot\underline{\overline{H}}^{*}}{2}-\frac{\underline{\overline{E}}\cdot\underline{\overline{D}}^{*}}{2}\right)\mathrm{d}v$$



$$\underline{P}_{\mathrm{cplx}}=-\oint_{\Sigma}\left(\underline{\overline{E}}\times\underline{\overline{H}}^{*}\right)\cdot\overline{n}\;\mathrm{d}A$$

Vectorul Poynting in complex

$$\underline{\overline{S}}=\underline{\overline{E}}\times\underline{\overline{H}}^{*}$$

$$\underline{\overline{E}}\cdot\underline{\overline{J}}^{*}=\left(\underline{E}_x\,\hat{x}+\underline{E}_y\,\hat{y}+\underline{E}_z\,\hat{z}\right)\cdot\left(\sigma\underline{E}_x\,\hat{x}+\sigma\underline{E}_y\,\hat{y}+\sigma\underline{E}_z\,\hat{z}\right)=\sigma\left(E_x^2+E_y^2+E_z^2\right)^2=\sigma\left(\widetilde{e_x^2}^{T}+\widetilde{e_y^2}^{T}+\widetilde{e_z^2}^{T}\right)=\sigma\,\widetilde{e^2}^{T}=\sigma\widetilde{(\overline{E}\cdot\overline{J})}^{T}=\widetilde{p_J}^{T}$$

$$\underline{\overline{E}}\cdot\underline{\overline{J}}^{*}=\widetilde{p_J}^{T}$$

Media pe o perioada a densitatii de putere transferata prin efect Joule (intr-un punct).

$$\frac{\underline{\overline{B}}\cdot\underline{\overline{H}}^{*}}{2}=\widetilde{w_m}^{T}$$

Media pe o perioada a densitatii de energie magnetica acumulata (intr-un punct).

$$\frac{\underline{\overline{E}}\cdot\underline{\overline{D}}^{*}}{2}=\widetilde{w_e}^{T}$$

Media pe o perioada a densitatii de energie electrica acumulata (intr-un punct).

$$\int_{D_\Sigma}\underline{\overline{E}}\cdot\underline{\overline{J}}^{*}\,\mathrm{d}v=\widetilde{P_J}^{T}$$

(global)

$$\int_{D_\Sigma}\frac{\underline{\overline{B}}\cdot\underline{\overline{H}}^{*}}{2}\,\mathrm{d}v=\widetilde{W_m}^{T}$$

(global)

$$\int_{D_\Sigma}\frac{\underline{\overline{E}}\cdot\underline{\overline{D}}^{*}}{2}\,\mathrm{d}v=\widetilde{W_e}^{T}$$

(global)

Local

$$-\operatorname{div}\underline{\overline{S}}=\widetilde{p_J}^{T}+2\,\mathrm{j}\,\omega\left(\widetilde{w_m}^{T}-\widetilde{w_e}^{T}\right)$$

Global

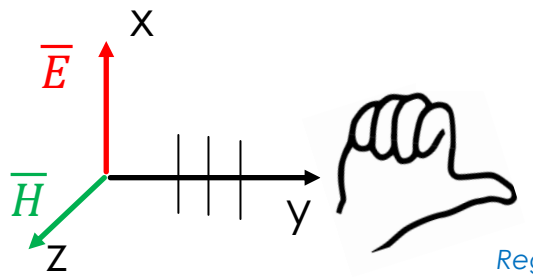
$$-\oint_{\Sigma}\left(\underline{\overline{E}}\times\underline{\overline{H}}^{*}\right)\cdot\overline{n}\;\mathrm{d}A=\widetilde{P_J}^{T}+2\,\mathrm{j}\,\omega\left(\widetilde{W_m}^{T}-\widetilde{W_e}^{T}\right)$$

$\underline{P}_{\mathrm{cplx}}=P+j\,Q$ $\quad P=\widetilde{P_J}^{T}$ Putere activa [W]

$\mathrm{Q}=2\,\omega\left(\widetilde{W_m}^{T}-\widetilde{W_e}^{T}\right)$ Putere reactiva [VAr]

5.7. Puterea transmisa

Unda plana polarizata liniar care se propaga dupa directia pozitiva a axei Oz (e o unda directa):



$\hat{x} \times \hat{y} = \hat{z}$

$$\overline{E}(z,t) = E_{xd}(z,t)\,\hat{x}$$

$$\overline{H}(z,t) = H_{xd}(z,t)\,\hat{y}$$

Regula mainii drepte poate fi folosita pentru a indica directia de transfer a puterii.

Vectorul Poynting

$$\overline{S}(z,t) = \overline{E} \times \overline{H} = E_{xd}(z,t)H_{xd}(z,t)\,\hat{z} = S(z,t)\hat{z}$$

Indica sensul transferului de energie.
Modulul lui reprezinta puterea transmisa de unda pe unitate de arie, este deci o **densitate de putere**. [W/m^2]

Intr-un mediu fara pierderi.

Campul electric si campul magnetic sunt in faza.
Nu exista atenuare. (In realitate, undele EM se atenueaza datorita raspandirii lor in spatiu, ele avand un front sferic si nu plan).
Impedanta caracteristica este un numar real.

$$Z_0 = \sqrt{\frac{\mu}{\varepsilon}} \qquad Z_0 = \frac{E_{xd}(z,t)}{H_{xd}(z,t)} \qquad \forall\, z, \forall t$$

Viteza de propagare a undei este $v = \frac{1}{\sqrt{\mu\varepsilon}}$

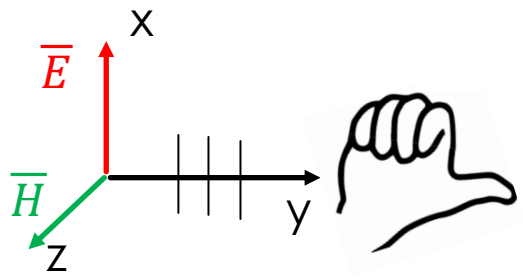
$$S(z,t) = \frac{E_{xd}^2(z,t)}{Z_0} = Z_0\, H_{xd}^2\,(z,t) = \sqrt{\frac{\varepsilon}{\mu}}\, E_{xd}^2(z,t) = \sqrt{\frac{\mu}{\varepsilon}}\, H_{xd}^2(z,t) = \frac{2}{\sqrt{\mu\varepsilon}}\frac{\varepsilon E_{xd}^2(z,t)}{2} = \frac{2}{\sqrt{\mu\varepsilon}}\frac{\mu\, H_{xd}^2(z,t)}{2} = 2\,v w_e = 2v w_m$$

$$w_e(z,t) = w_m(z,t) \quad \Rightarrow \quad w(z,t) = 2w_e(z,t) = 2w_m(z,t) \quad \Rightarrow \quad S(z,t) = v\, w(z,t)$$

Unda plana, intr-un mediu fara pierderi, e tot timpul la rezonanta.

Densitatea de energie totala.

5.7. Puterea transmisa



$\hat{x} \times \hat{y} = \hat{z}$

Regula mainii drepte poate fi folosita pentru a indica directia de transfer a puterii.

Intr-un mediu cu pierderi.

Rationamentul se face in r.a.p.

$\underline{\overline{S}} = \underline{\overline{E}} \times \underline{\overline{H}}^*$ Marimile depind de z.

$$\underline{\overline{E}}(z) = \underline{E}_0\, e^{-\underline{\gamma} z}\, \hat{x} = \underline{Z}_c \underline{H}_0\, e^{-\underline{\gamma} z}\, \hat{x}$$

$$\underline{\overline{H}}(z) = \frac{\underline{E}_0}{\underline{Z}_c}\, e^{-\underline{\gamma} z}\, \hat{y} \quad \Rightarrow \quad \underline{\overline{H}}^*(z) = \frac{\underline{E}_0^*}{\underline{Z}_c^*}\, e^{-\underline{\gamma}^* z}\, \hat{y} = \underline{H}_0^*\, e^{-\underline{\gamma}^* z}\, \hat{y}$$

$$\underline{\overline{S}} = \underline{\overline{E}} \times \underline{\overline{H}}^* = \underline{E}_0\, e^{-\underline{\gamma} z}\, \frac{\underline{E}_0^*}{\underline{Z}_c^*}\, e^{-\underline{\gamma}^* z}\, \hat{k} = \underline{S}\hat{k} \qquad \underline{\overline{S}} = \underline{\overline{E}} \times \underline{\overline{H}}^* = \underline{Z}_c \underline{H}_0 e^{-\underline{\gamma} z}\, \underline{H}_0^*\, e^{-\underline{\gamma}^* z}\, \hat{k} = \underline{S}\hat{k}$$

$$\underline{S}(z) = \frac{\underline{E}_0 \underline{E}_0^*}{\underline{Z}_c^*} e^{-\left(\underline{\gamma}+\underline{\gamma}^*\right) z} = \frac{E_0^2}{\underline{Z}_c^*} e^{-2\,\alpha z} \qquad \underline{S}(z) = \underline{Z}_c \underline{H}_0 \underline{H}_0^* e^{-\left(\underline{\gamma}+\underline{\gamma}^*\right) z} = \underline{Z}_c\, H_0^2\, e^{-2\,\alpha z}$$

Pentru a afla media pe o perioada a densitatii de putere transmise:

$$\widetilde{S(z,t)}^T = \frac{1}{T}\int_0^T S(z,t)\, \mathrm{d}t = \mathrm{Real}\left(\underline{S}(z)\right)$$

5.7. Puterea transmisa – exemple (1) [Niknejdad07]

1) Campul electric (unda sferica) produs de o statie de baza dintr-o retea de telefonie mobila.

Sa presupunem ca o statie de baza dintr-o retea de telefonie mobila are o putere de emisie de P=10 kW. Vom estima valoarea maxima a intensitatii campului electric la o distanta de 1 metru de antena (de fapt valorea medie a acestei marimi pe sfera de raza 1metru).

Pp. ca mediul este vid (fara pierderi, de impedanta $Z_0 = 377\ \Omega$) si presupunem ca sursa emite izotrop (la fel in toate directiile), produce o unda sferica. In aceste conditii puterea transmisa de antena trebuie sa se regaseasca pe o sfera de raza r, centrata in antena. Fiind unda sferica, directia de propagare este radiala, si datorita simetriei, valoarea densitatii de putere este aceeasi pe sfera.

$$P = P(r) = -\oint_{\Sigma} \overline{S} \cdot \overline{n}\ \mathrm{d}A = \mathrm{S_{mediu}(r)}\ 4\,\pi r^2 \quad \Rightarrow \quad S_{\mathrm{mediu}}(r) = S(r) = \frac{P}{4\,\pi r^2} \quad \left[\frac{W}{m^2}\right]$$



Aproximand local unda sferica cu o unda plana, putem estima modulul intensitatii campului electric

Atunci

$$S = \frac{E^2}{Z_0}$$

unde E este valoare efectiva $\Rightarrow$ $S(r) = \frac{E_{\max}^2(r)}{2\,Z_0}$ $\Rightarrow$ $E_{\max}(r) = \sqrt{2\,Z_0\,S(r)} = \sqrt{Z_0\,\frac{P}{2\,\pi r^2}}$

$$E_{\max}(1) = \sqrt{377\,\frac{10^4}{2\,\pi}} = 775 \ \text{[V/m]} \qquad E_{\max}(1000) = \sqrt{377\,\frac{10^4}{2\,\pi\,10^6}} = 775 \ \text{[mV/m]}$$

5.7. Puterea transmisa – exemple (2) [Niknejdad07]

2) Campul electric produs de un telefon mobil.

Sa presupunem ca un telefon mobil are o putere de emisie de P = 1W. Vom estima valoarea maxima a intensitatii campului electric la o distanta de 10 cm de telefon (de fapt valorea medie a acestei marimi pe sfera de raza 10 cm).

Am putea sa reluam ipotezele si calculul din exemplu anterior. Rezultatul ar fi

$$E_{\max}(r) = \sqrt{2\, Z_0\, S(r)} = \sqrt{Z_0 \frac{P}{2\,\pi r^2}}$$

$$E_{\max}(0.1) = \sqrt{377\ \frac{1}{2\ \pi\ 0.01}} = 77.5 \quad \text{[V/m]}$$

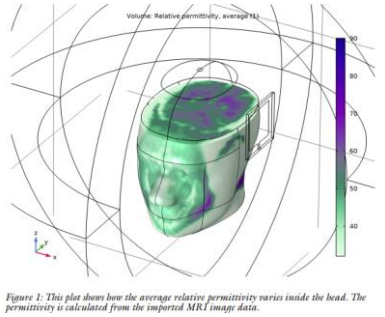


Figure 1: This plot shows how the average relative permittivity varies inside the head. The permittivity is calculated from the imported MRI image data.

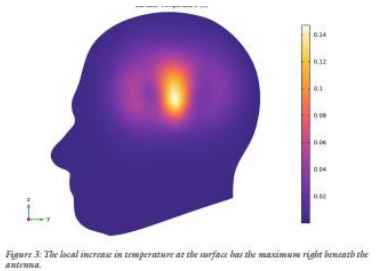


Figure 3: The local increase in temperature at the surface has the maximum right beneath the antenna.

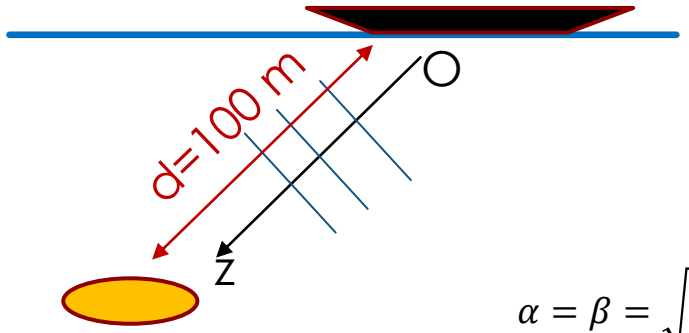
https://www.comsol.com/model/download/1093691/models.rf.sar_in_human_head.pdf

In realitate un astfel de model nu este potrivit la o distanta mica de antena, nu putem presupune mediul omogen si fara pierderi, nici aproximatia de unda plana nu este potrivita.
Pentru a calcula campul EM intr-un astfel de scenario, trebuie folosite model numerice care sa include tipul de radiatie al antenei si forma si proprietatile EM ale capului.

5.7. Puterea transmisa – exemple (3) [Niknejdad07]

3) Comunicarea dintre un vapor si un submarin.

Sa presupunem ca un submarin se afla scufundat la 100 m fata de un vapor aflat pe mare. Personalul aflat pe vapor doreste sa comunice cu submarinul, folosind frecventa $f = 3\ \text{kHz}$. Sa se calculeze puterea necesara emitatorului (pe vapor) astfel inca receptorul (submarin) sa primeasca un semnal cu o amplitudine de minimum $E_{\min} = 1\mu\text{V/m}$.



Proprietatile apei de mare

$\sigma = 4\ \text{S/m},\ \varepsilon_r = 81, \mu_r = 1$

Este necesara folosirea unor frecvente mici pentru comunicare deoarece apa marii este conductoare. Tangenta unghiului de pierderi este

$$\text{tg}\,\delta_c = \frac{\sigma}{\omega\,\varepsilon} \approx 10^5 \gg 1$$

Apa marii poate fi considerate foarte bun conductor, chiar si la aceasta frecventa forte mica.

Vom folosi formulele demonstrate la conductoarele foarte bune (sectiunea 5.3.).

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} \approx 0.2 \qquad Z_c = \sqrt{\frac{\omega\mu}{\sigma}} \approx 0.08\ \Omega \qquad \theta = 45^o$$

$$\lambda = \frac{2\,\pi}{\beta} = 29\ \text{m} \qquad E(z,t) = E_0\sqrt{2}e^{-\alpha z}\sin(\omega t - \beta z)$$

$$\delta = \frac{1}{\alpha} = 4.6\ \text{m} \qquad H(z,t) = \frac{E_0}{Z_c}\sqrt{2}e^{-\alpha z}\sin(\omega t - \beta z - \theta)$$

La receptor: $E_0 e^{-\alpha d}\sqrt{2} \geq E_{\min}$ $\qquad E_0 \geq \frac{E_{\min}}{\sqrt{2}} e^{\alpha d}$

La sursa: $\underline{S}(0) = \frac{E_0^2}{\underline{Z_c^*}} = \frac{E_0^2}{Z_c} e^{j\theta}$

$$P(0) = \text{Real}\left(S(0)\right) \geq \frac{E_{\min}^2\, e^{2\alpha d}\cos\theta}{2\,Z_c} = 37\ \frac{\text{MW}}{\text{m}^2}$$

Enorm!

La receptor puterea este: $\underline{S}(d) = \frac{E_0^2}{\underline{Z_c^*}} e^{-2\alpha d}$

$$P(d) = \text{Real}\left(S(d)\right) = P(0)e^{-2\alpha d} = 4.5 \cdot 10^{-12}\ \frac{\text{W}}{\text{m}^2} = 4.5\ \frac{\text{pW}}{\text{m}^2}$$

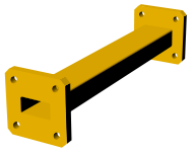
Cod matlab disponibil

La d = 10 m, $P(0) = 0.35\frac{\text{nW}}{\text{m}^2}$, $P(d) = 4.5\frac{\text{pW}}{\text{m}^2}$

5.8. Ghiduri de unda (facultativ)

- **Ghid de unda** = sistem electromagnetic cu structura omogena, care determina propagarea undei intr-o directive determinate, mentinand un anumit grad de concentrare a campului EM in plane perpendicular pe directia de propagare.
- **Ghid de unda metalic** – este te alcatuit dintr-un conductor tubular care reflecta unda EM si separa domeniul dielectric interior de exterior, astfel incat unda se propaga numai in dielectricul interior.

Sunt folosite in excitarea antenelor, radare, cuptoare cu microunde.

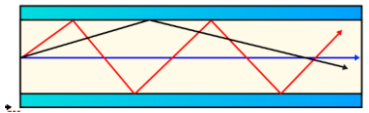


https://en.wikipedia.org/wiki/Waveguide_(radio_frequency)

	Linii electrice	Ghiduri de unda
Moduri de propagare	Numai TEM	TE, TM, hibrid
Frecvente de lucru	De la 0 la o frecventa maxima limitata de efectul pellicular in conductoare si pierderile in dielectrici.	Pot fi folosite numai peste o anumita frecventa, numita frecventa de taiere (cut-off frequency)
Constructiv	Au minim 2 conductoare	Au un conductor sau niciunul.

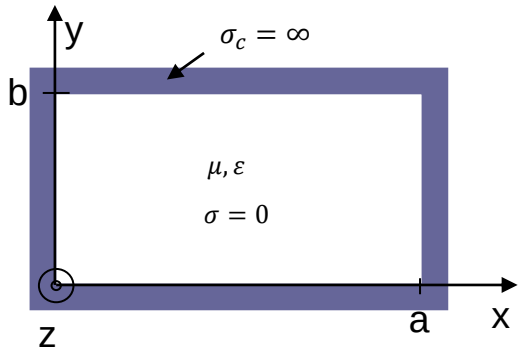
Ghidul de unda dielectric
(fibra optica, cu reflexie totala, deci fara pierderi prin refractie, avand si 100km.)

https://en.wikipedia.org/wiki/Waveguide_(optics)



5.8. Ghiduri de unda (facultativ)

Ecuatiile de ordinul al II-lea – ghid de unda in regim armonic permanent



Mediul dielectric interior:
- izotrop
- liniar
- omogen
- perfect izolant
- fara surse interne de camp

Conductorul ghidului
- perfect conductor

$$\begin{cases} \text{rot}\,\mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \text{rot}\,\mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t} \\ \text{div}\,\mathbf{D} = \rho \\ \text{div}\mathbf{B} = 0 \\ \mathbf{D} = \overline{\varepsilon}\mathbf{E} + \mathbf{P}_\text{p} \\ \mathbf{B} = \overline{\mu}\mathbf{H} + \mu_0\mathbf{M}_p \\ \mathbf{J} = \overline{\sigma}\mathbf{E} + \mathbf{J}_i \end{cases} \Rightarrow \begin{cases} \text{rot}\,\underline{\mathbf{E}} = -j\omega\mu\,\underline{\mathbf{H}} \\ \text{rot}\,\underline{\mathbf{H}} = j\omega\varepsilon\underline{\mathbf{E}} \\ \text{div}\,\varepsilon\underline{\mathbf{E}} = 0 \\ \text{div}\,\mu\underline{\mathbf{H}} = 0 \end{cases}$$

$$\Rightarrow \text{rot rot}\underline{\mathbf{E}} = -j\omega\mu\text{rot}\underline{\mathbf{H}}$$

$$\text{grad div}\underline{\mathbf{E}} - \Delta\underline{\mathbf{E}} = \omega^2\mu\varepsilon\underline{\mathbf{E}}$$

$$\Delta\underline{\mathbf{E}} + k^2\,\underline{\mathbf{E}} = 0$$

In mod similar: $\Delta\underline{\mathbf{H}} + k^2\,\underline{\mathbf{H}} = 0$

Constanta de propagare:

$$k^2 = \omega^2\mu\varepsilon = \frac{(2\pi f)^2}{(\frac{1}{\sqrt{\mu\varepsilon}})^2} = \left(\frac{2\pi}{\frac{v}{f}}\right)^2 = \left(\frac{2\pi}{\lambda}\right)^2 \Rightarrow$$

In vid: $k_0^2 = \omega^2\mu_0\varepsilon_0$

Vom pp $a \geq b$ (altfel renotam dimensiunile)

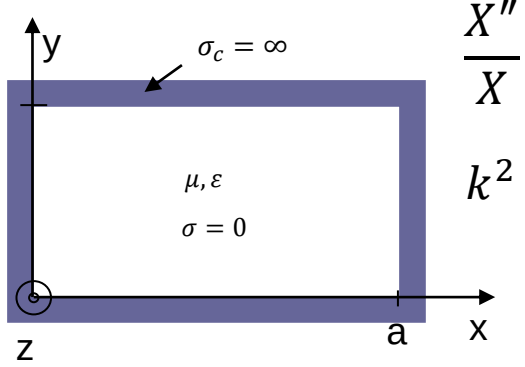
$$k^2 = k_0^2\mu_r\varepsilon_r = k_0^2 n^2\,, \qquad n = \sqrt{\mu_r\varepsilon_r} = \frac{c}{v}$$

Indice de refractie al mediului

5.8. Ghiduri de unda (facultativ)

Ecuatiile de ordinul al II-lea – ghid de unda in regim armonic permanent

$\Delta\underline{\mathbf{E}} + K^2\,\underline{\mathbf{E}} = 0$ Metoda separarii variabilelor $\underline{E}_z(x,y,z) = CX(x)Y(y)Z(z)$



$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = -k^2 \Rightarrow \frac{X''}{X} = -k_x^2 \qquad \frac{Y''}{Y} = -k_y^2 \qquad \frac{Z''}{Z} = -k_z^2 \Rightarrow$$

$$k^2 = \underbrace{k_x^2 + k_y^2}_{k_t^2} + k_z^2 \Rightarrow k^2 = k_t^2 + k_z^2 \Rightarrow k_z = \sqrt{k^2 - k_t^2} \Rightarrow$$

k_t^2 - constanta de separare transversala

Daca dorim propagare dupa Oz: $\frac{Z''}{Z} = -k_z^2$ ec. caracteristica $r^2 = -k_z^2 \Rightarrow r = jk_z \Rightarrow k_z = \text{real} \Rightarrow$

$$k^2 > k_t^2 \Rightarrow \omega^2\mu\varepsilon > K_t^2 \Rightarrow \omega > \frac{K_t}{\sqrt{\mu\varepsilon}} = \omega_c = \text{frecventa de taiere}$$

! Ghidurile de unda pot folosi la propagarea semnalelor doar pentru frecvente mai mari decat frecventa de taiere

5.8. Ghiduri de unda (facultativ)

Ecuatiile de ordinul al II-lea – ghid de unda in regim armonic permanent

Se demonstreaza ca solutia generala a ghidului de unda dreptunghiular are:

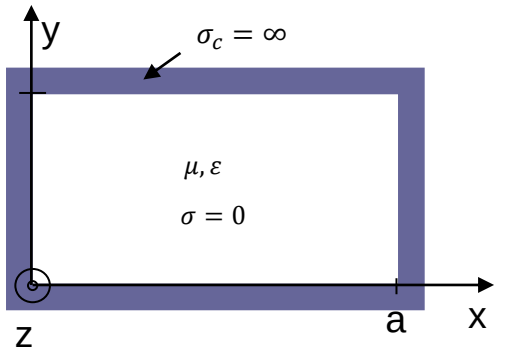
- componentele longitudinale:

$$\underline{E_z}(x,y,z) = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y)e^{-jk_z z}$$

$$\underline{H_z}(x,y,z) = (B_1 \cos k_x x + B_2 \sin k_x x)(B_3 \cos k_y y + B_4 \sin K_y y)e^{-jk_z z}$$

- componentele transversale, in functie de componentele longitudinale:

$$\underline{E_x}(x,y,z) = -\frac{1}{k_t^2}\left(k_z \frac{\partial \underline{E_z}}{\partial x} + j\omega\mu \frac{\partial \underline{H_z}}{\partial y}\right), \quad \text{etc.}$$



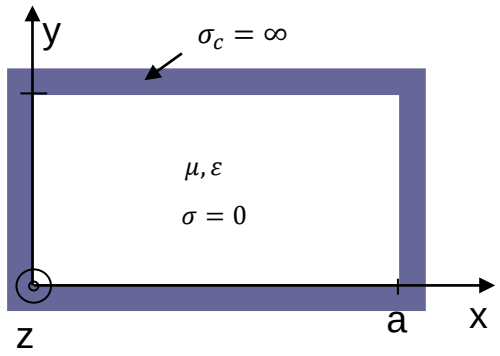
Ele pot determina diferite configuratii posibile ale campului electromagnetic, numite **moduri**. Exista 4 tipuri de moduri:

TEM ($E_x = 0$, $H_x = 0$) – nu este posibil pentru ghiduri de unda
TM ($E_x \neq 0$, $H_x = 0$)
TE ($E_x = 0$, $H_x \neq 0$)
HEM ($E_x \neq 0$, $H_x \neq 0$)

5.8. Ghiduri de unda (facultativ)

Modul TE ($E_z = 0, H_z \neq 0$)

$$\underline{H_z}(x,y,z) = (B_1 \cos k_x x + B_2 \sin k_x x)(B_3 \cos k_y y + B_4 \sin K_y y)e^{-jk_z z}$$



Din conditia de frontiera $E_t = 0$

$$\underline{H_z}(x,y,z) = (B_1 \cos k_x x)(B_3 \cos k_y y)e^{-jk_z z}$$

$$\Rightarrow \begin{cases} \sin k_x a = 0 \Rightarrow k_x a = m\pi \Rightarrow k_x = \dfrac{m\pi}{a} \\ \sin k_y b = 0 \Rightarrow k_y b = n\pi \Rightarrow k_y = \dfrac{n\pi}{b} \end{cases} \Rightarrow \text{Mod } TE_{m,n}$$

Frecventa de taiere:

$$\omega_c = \frac{k_t}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{v}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$k_t^2 = k_x^2 + k_y^2$$

Deoarece $a > b$ ➡ **Modul de propagare cu cea mai mica frecventa de taiere este $TE_{1,0}$:**

$$m = 1 \Rightarrow k_x = \frac{\pi}{a} \qquad n = 0 \Rightarrow k_y = 0$$

$$k_z = j\,\beta = j\,\omega\sqrt{\mu\varepsilon}\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} \quad \omega > \omega_c$$

Viteza de faza $v = \frac{\omega}{\beta} = \frac{c}{\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2}} > c$ **e mai mare ca viteza luminii**

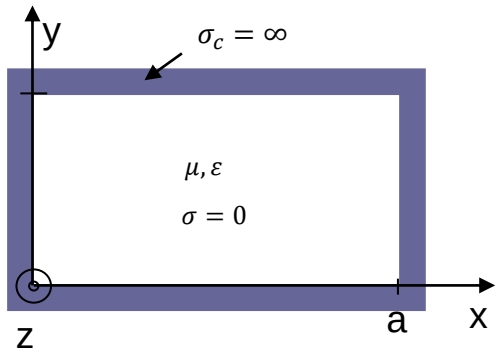
Viteza de grup $v_g = \frac{d\omega}{d\beta} = c\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} < c$

Aceasta da viteza de transfer a datelor, tinde catre zero cand ω tinde catre frecventa de taiere. La frecventa de taiere ghidul nu mai poate transfera date.

5.8. Ghiduri de unda (facultativ)

Modul TM ($H_z = 0, E_z \neq 0$)

$$\underline{E_z}(x,y,z) = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y)e^{-jk_z z}$$



Din conditia de frontiera $E_t = 0$

$$\underline{E_z}(x,y,z) = (A_2 \sin k_x x)(A_4 \sin k_y y)e^{-jk_z z}$$

$$\begin{cases} \sin k_x a = 0 \Rightarrow k_x a = m\pi \Rightarrow k_x = \dfrac{m\pi}{a} \\ \sin k_y b = 0 \Rightarrow k_y b = n\pi \Rightarrow k_y = \dfrac{n\pi}{b} \end{cases} \Rightarrow \text{Mod } TM_{m,n}$$

Frecventa de taiere:

$$\omega_c = \frac{k_t}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{v}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$k_t^2 = k_x^2 + k_y^2$$

Deoarece $a > b$ ➡ **Modul de propagare cu cea mai mica frecventa de taiere este $TM_{1,1}$:**

$$m = 1 \Rightarrow k_x = \frac{\pi}{a} \qquad n = 1 \Rightarrow k_y = \frac{\pi}{b}$$

$$k_z = j\,\beta = j\,\omega\sqrt{\mu\varepsilon}\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} \quad \omega > \omega_c$$

Viteza de faza $v = \dfrac{\omega}{\beta} = \dfrac{c}{\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2}} > c$ **e mai mare ca viteza luminii**

Viteza de grup $v_g = \dfrac{d\omega}{d\beta} = c\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} < c$

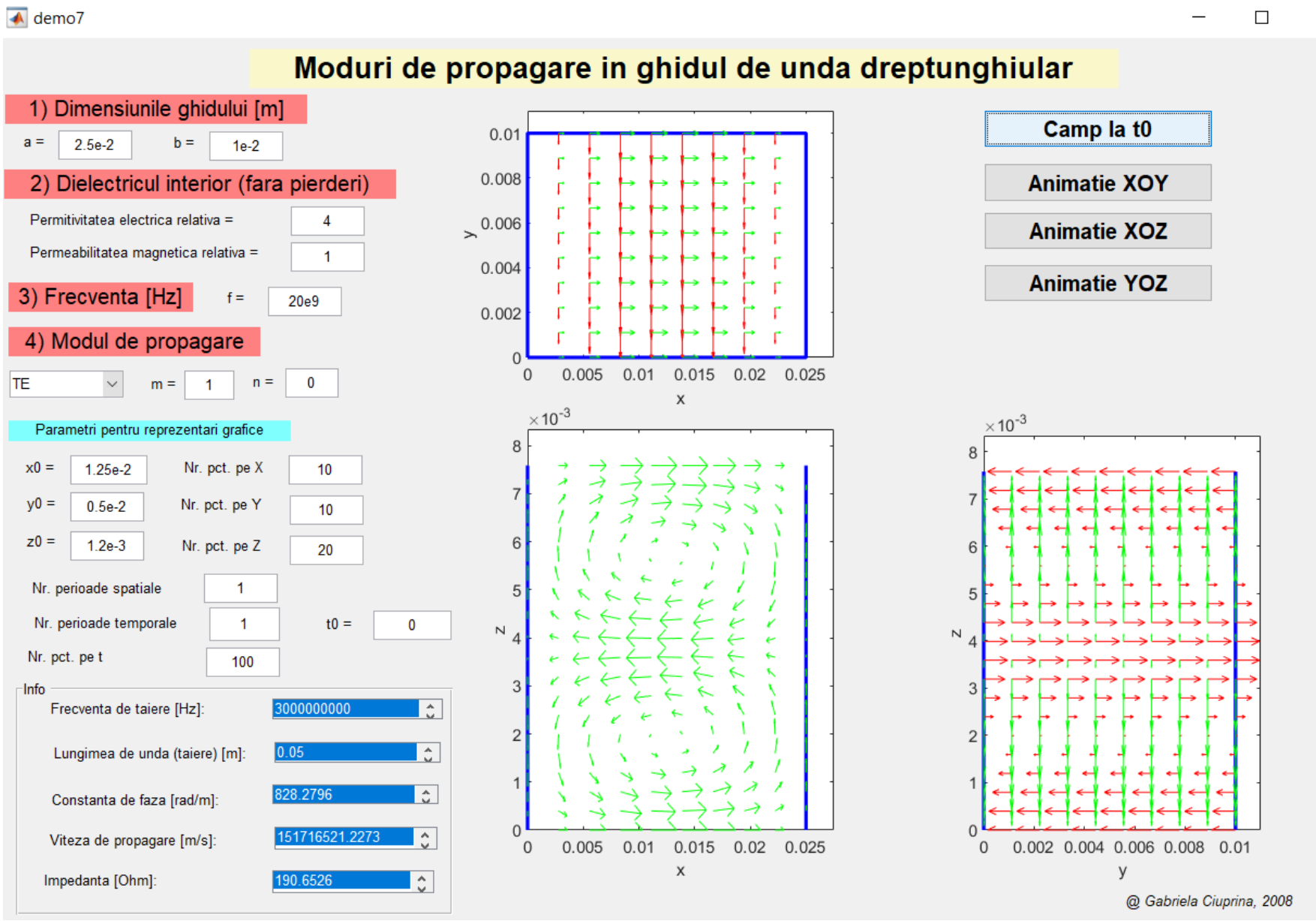
Aceasta da viteza de transfer a datelor, tinde catre zero cand ω tinde catre frecventa de taiere. La frecventa de taiere ghidul nu mai poate transfera date.

5.8. Ghiduri de unda

TE_{10}

Matlab demo7.m

Indicii reprezinta numarul de semiperioade dupa Ox si Oy: aici o semiperioada dupa Ox, constant dupa Oy – nu depinde de y.

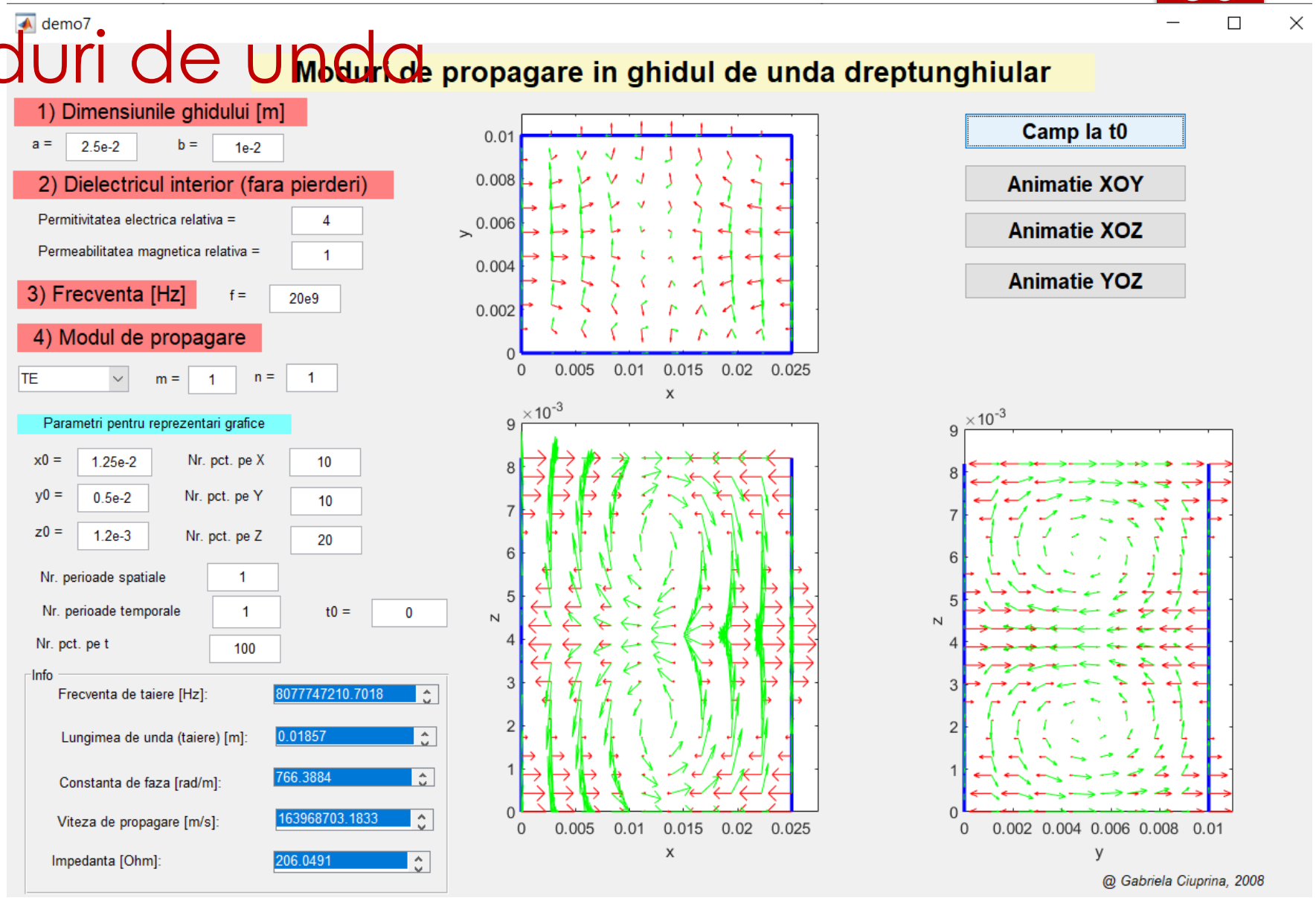


5.8. Ghiduri de unda

TE_{11}

Matlab demo7.m

1 semiperioada dupa Ox
1 semiperioada dupa Oy

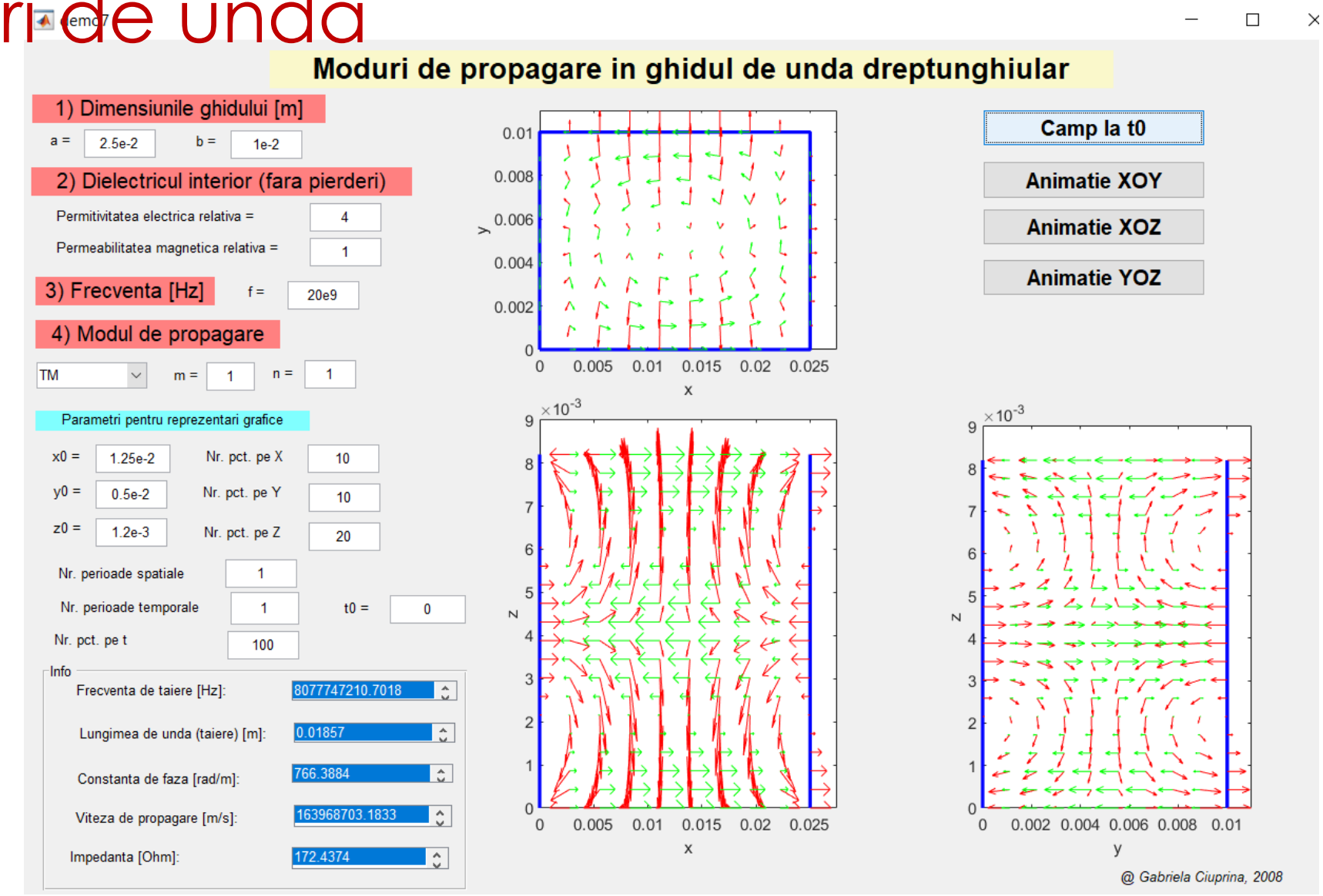


5.8. Ghiduri de unda

TM_{11}

Matlab demo7.m

1 semiperioada dupa Ox
1 semiperioada dupa Oy

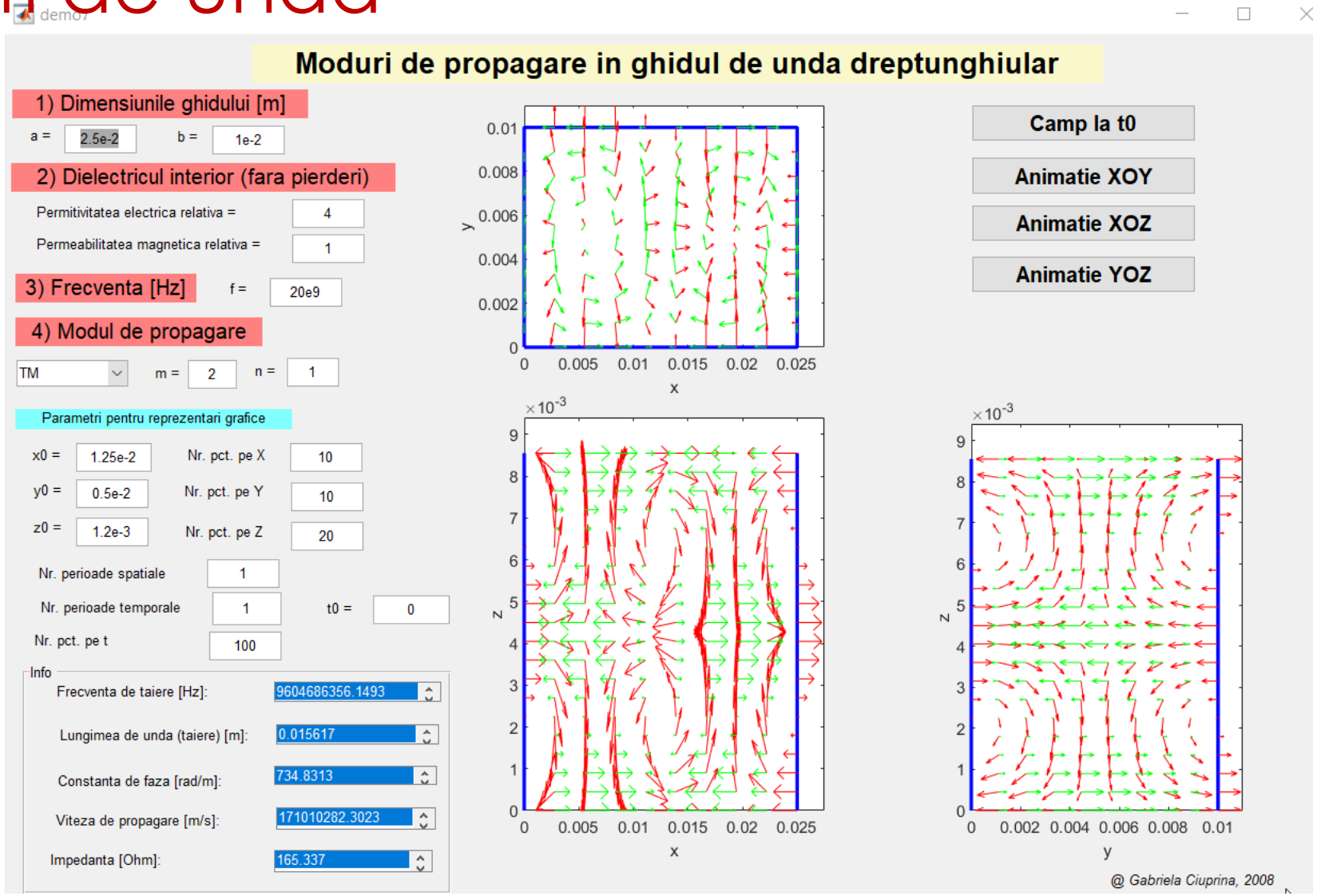


5.8. Ghiduri de unda

TM_{21}

Matlab demo7.m

2 semiperioade(o perioada) dupa Ox
1 semiperioada dupa Oy

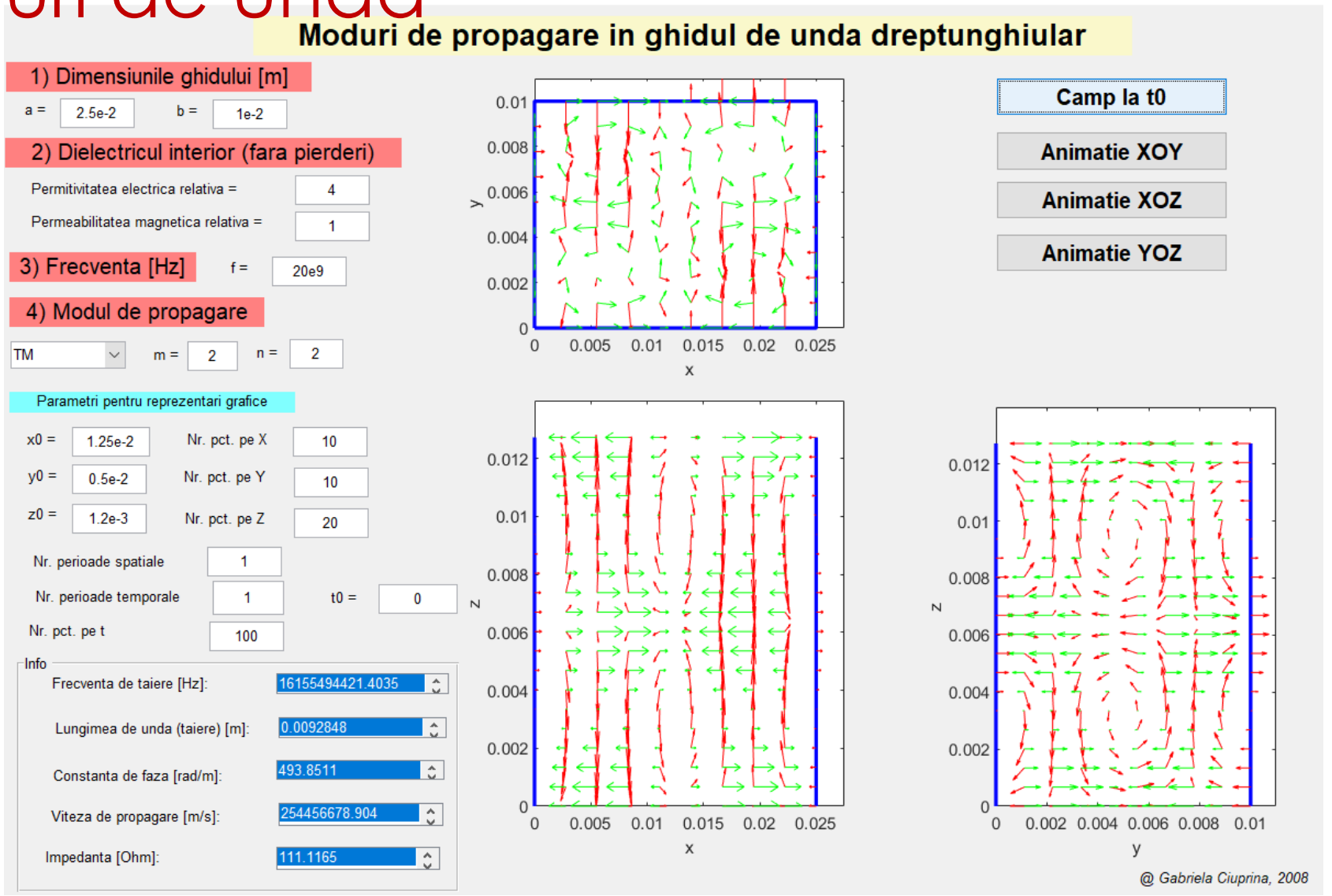


5.8. Ghiduri de unda

TM_{22}

Matlab demo7.m

2 semiperioade (o perioada) dupa Ox
2 semiperioade (o perioada) dupa Oy



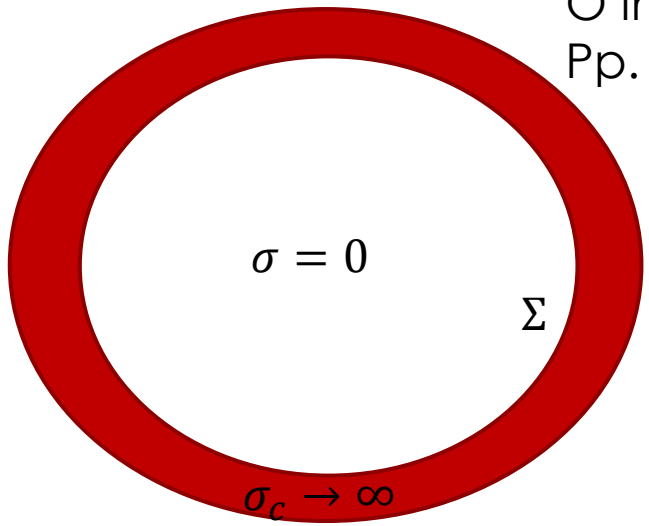
5.9 Sisteme rezonante (facultativ)

- Un sistem electromagnetic functioneaza la **rezonanta** daca, in regim armonic permanent puterea reactiva primita de sistem este nula.

$$Q = 0$$

Aplicatii: cuptorul cu microunde, lasere, acceleratoare de particule, generatoare si oscilatoare de microunde.

Ideea: cavitatea rezonanta cu pereti PEC



O incinta perfect izolanta inconjurata de un material perfect conductor.

Pp. r.a.p. $\sigma = 0 \Rightarrow \bar{J} = 0 \Rightarrow P = 0$

$$\sigma_c \to \infty \Rightarrow \overline{E} = 0 \Rightarrow \underline{P}_{\text{cplx}} = -\oint_{\Sigma} \left(\underline{\overline{E}} \times \underline{\overline{H}}^{*}\right) \cdot \overline{n}\, \mathrm{d}A = 0$$

$$\underline{P}_{\text{cplx}} = P + j\, Q$$

$$Q = 0$$

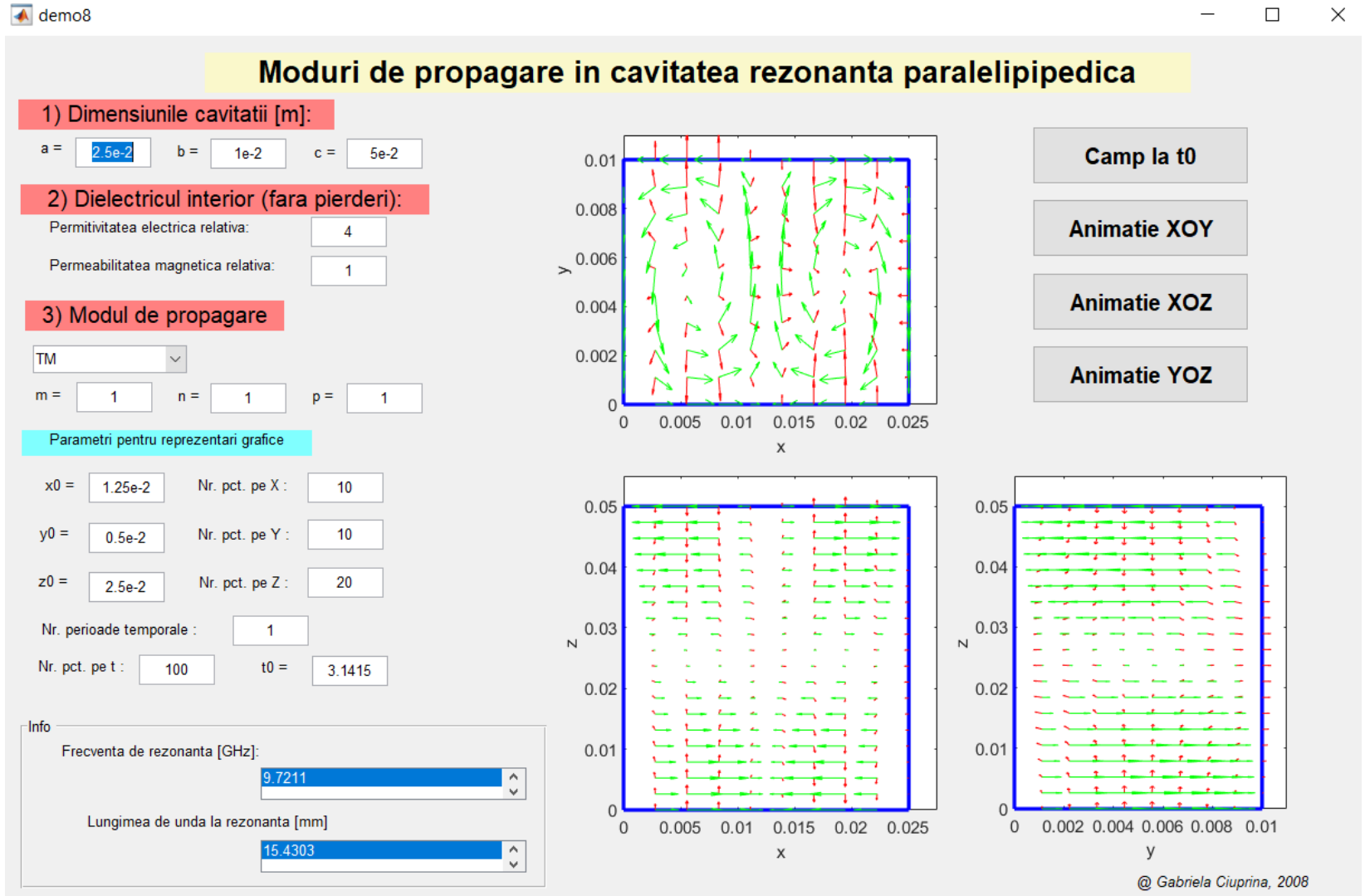
$$Q = 2\,\omega \left(\widetilde{W_m}^{T} - \widetilde{W_e}^{T}\right) \Rightarrow \widetilde{W_m}^{T} = \widetilde{W_e}^{T}$$

O cavitate rezonanta se poate obtine inchizand la ambele capete un ghid de unda.

5.9 Sisteme rezonante

TM_{111}

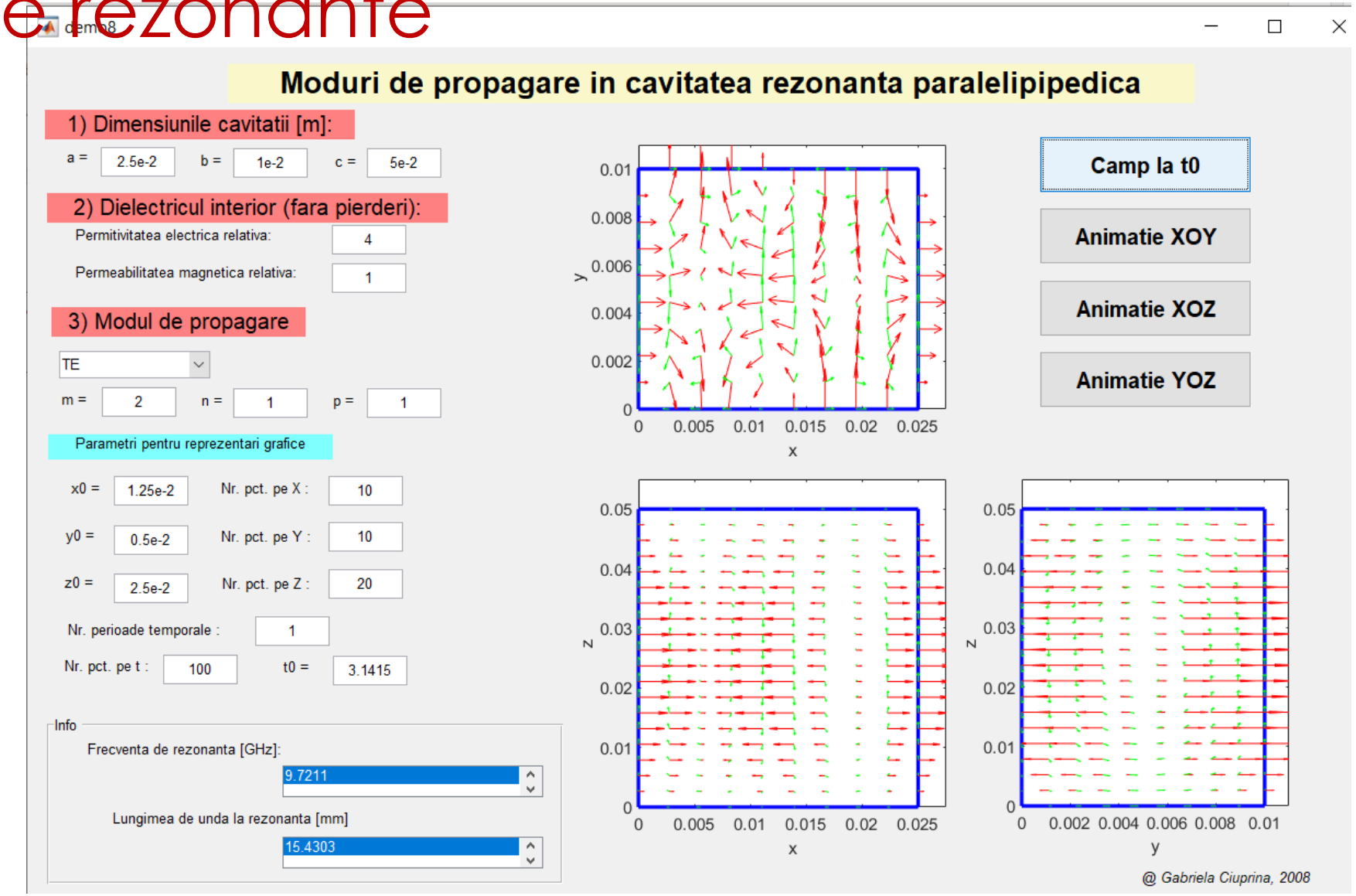
Matlab demo8.m



5.9 Sisteme rezonante

TE_{211}

Matlab demo8.m



Concluzii

- O unda plana este o unda TEM care depinde de o singura coordonata spatiala carteziana.
- Undele plane sunt idealizari cu front plant ale undelor EM, folosite pentru rationamente la distante mari de sursele de radiatie (antene).
- Ecuatiile undelor plane, dar si solutiile lor, sunt similare cu ecuatiile liniilor, in care nu exista pierderi in conductoare ($R_l = 0$) si in loc de u [V], i [A], C_l [F/m], L_l [H/m], G_l [S/m] apar E_x[V/m], H_y [A/m], ε [F/m], μ [H/m], σ [S/m]. Noutatea consta in polarizarea undelor, care poate fi liniara, circulara sau eliptica.
- Cand se reflecta/refracta, undele EM respecta legille opticii geometrice.
- In cazul in care undele nu sunt alcatuite din semnale de o singura frecventa (monocromatice), atunci rationamentele se pot face prin superpozitie. In medii dispersive, propagarea componentelor nu se face cu aceeasi frecventa, semnalele se propaga distorsionat. Intr-un astfel de scenariu, este utila si definirea conceptului de viteza de grup, nu numai a vitezei de faza (concept folosit in acest capitol).

Referinte

A. Tomescu si F.M.G Tomescu, Sisteme cu Microunde, Editura MatrixROM (o pot imprumuta celor interesati)
A. Tomescu si F.M.G Tomescu, Sisteme cu Microunde, Editura MatrixROM (o pot imprumuta celor interesati)

Sophocles J. Orfanidis, Electromagnetic Waves and Antennas, 2016 disponibila online la https://www.ece.rutgers.edu/~orfanidi/ewa/ (cap 2, 5 si 7)

Jin Au Kong – Electromagnetic Wave Theory – EMW Publishing, disponibila la

https://www.piers.org/piers/files/BOOK_ElectromagneticWaveTheory_by_JINAUKONG_FreeRelease2021.pdf

Ali M. Niknejad, Electromagnetics for High-Speed Analog and Digital Communication Circuits, Cambridge 2007 (o pot imprumuta celor interesati)